Uncovering patterns of semantic predictability in sentence processing

Cassandra L. Jacobs[1], Ryan J. Hubbard[23], Loïc Grobol[4], & Kara D. Federmeier[2]
1: State University of New York at Buffalo
2: University of Illinois at Urbana-Champaign
3: State University of New York at Albany
4: Université Paris Nanterre

Corresponding author contact information:

Cassandra Jacobs
614 Baldy Hall
Buffalo, NY 14260
cxjacobs@buffalo.edu

Abstract

Psycholinguistic researchers often collect cloze probabilities in order to measure the predictability of upcoming words but have largely discarded the variability in the structure of responses people provide. This variability in the semantic structure of responses may be important for understanding selection during language production; however, it has proven difficult to model the semantic variability of participants' responses, and thus upcoming semantic uncertainty. Recent advances in large language models (LLMs) permit us to approximate the degree of semantic variability in cloze responses, but most methods are restricted to symbolic or hand-crafted meaning representations. We show in two studies that Bayesian Gaussian mixture models can cluster LLM representations of participants' responses and produce coherent, taxonomically similar clusters. We apply these clustering algorithms to response time data in a serial cloze task and show that the semantic structure of cloze responses influences how quickly people are able to provide a response. We show clear effects of semantic competition on production speed. In addition to providing novel operationalizations of what semantic competition might look like in the cloze task, we explain how this clustering method is extensible to other datasets and applications of interest to researchers of semantic processing in psycholinguistics.

Introduction

A popularly held belief is that true friends can always finish each other's … sandwiches. The remarkable ability for us to predict the continuations of what others will say is a hallmark of human language comprehension. Indeed, when people are given the task to guess the next word of a sentence or to fill in the blank in a cloze task (Taylor, 1953), people often guess words that are related to one other. Despite this systematicity, what we anticipate others will say is not perfectly predictable from the context, and it is not always obvious to determine how closely related participants' responses generally are. A word-centric focus in the use of cloze probability data may obscure the degree to which the future may be semantically, versus lexically, constrained. Thus, a major question in psycholinguistic research is how semantic information may be easier to predict than wordform information, and the degree to which semantic information facilitates production or creates competition. However, characterizing the semantic structure of cloze responses and organizing them into classes of related words is difficult to do by hand. Similarly, research in language production seeks to understand how the mechanisms that enable lexical selection are affected by the availability of alternatives. A long-standing debate in language production concerns whether different potential responses directly compete with each other for selection, or whether producers simply select the word that reaches a threshold of activation first (Levelt, 1989; Mahon et al., 2007; Oppenheim, 2023; Spalek, Damian, & Bölte, 2013; Staub et al., 2015).

A major challenge for modeling semantic processing in language comprehension and production is in representing the semantics of words and their relationship to each other (Landauer & Dumais, 1997). In this work, we approach the question of representing a word's meaning by approximating that meaning using a distributional semantics approach that we apply over data collected from cloze tasks. We combine neural language model representations with Bayesian Gaussian mixture modeling methods for clustering cloze responses to a variety of written English sentences. The resulting clusters act as a proxy for the semantic factors that readers and producers may be sensitive to, and we apply probabilistic measures of semantic predictability derived from clustering behavior to assess the contribution of semantics to the speed of production. We focus on cloze data for two reasons. First, cloze probabilities have been central to both our early and current understanding of the relationship between predictability and comprehension measures, including reading times and neural responses to written words (Kutas & Hillyard, 1980; see de Varda et al., 2023 for a review). Cloze data are also revealing about language production (Staub et al., 2015). For example, cloze responses are systematically biased "away" from word co-occurrence statistics by being composed of words that are more frequent, concrete, and semantically similar to the context than would be expected by n-gram statistics alone (Smith & Levy, 2011). We first review the cloze task as it has been applied in comprehension and production research and outline our motivation for clustering large language model-based representations of cloze responses to better understand semantic processing in production.

The cloze task and large language models (LLMs)

In the cloze task (Taylor, 1953), a person reads or listens to a sentence with a missing word or words, and they must then produce the word(s) to best continue or complete the sentence. Often, the sentence fragments used in the task are designed to yield words with differing levels of predictability. For example, some contexts are highly constraining towards a certain response, such as:

1. "At night the old woman locked the ____."

Here, most (more than 90%) of the participants tend to produce the word "*door"*. Other contexts, however, may be less constraining:

2. "She cleaned the dirt from her _____."

In this case, only about a third of the people produce the same word ("*shoes*") as the most likely completion.

In cloze tasks, participants may be asked to fill in one or multiple blanks. The to-be-guessed words may be presented serially (or "cumulatively"; Luke & Christianson, 2016; Lowder, Choi, Ferreira, & Henderson, 2017), with words being added to the sentence one at a time after each participant's response. Alternately, participants may read a surrounding context with one or more missing words. Participants typically provide one (but occasionally more than one; Roland et al., 2012) response as to what they believe is the most likely way to complete the sentence, the best word in the context, or the first word that comes to mind.

Producing responses in the cloze task is thought to engage similar prediction mechanisms as in language comprehension (Dell & Chang, 2014), such as those that facilitate self-paced reading. That is, both reading tasks and the cloze task require integrating the current words of the context in order to generate a possible unknown future linguistic outcome, such as the next word or syntactic structure in a sentence. The link between producing cloze responses and language comprehension is typically quantified by assessing the *probability* of the upcoming word (or the word currently being processed) in a sentence using the aggregated cloze responses of many participants – e.g., cloze probabilities of 94% versus 31% for *door* versus *shoes* in the prior examples. Cloze probabilities are then used as predictors or correlates of reading (or other response) times and/or neural signals. There is a robust tendency for high-probability words to be read faster (de Varda et al., 2023; Luke & Christianson, 2016; Shain et al., 2024). Cloze probabilities are also good predictors of a neural signature related to semantic processing – i.e., the N400 (Kutas & Hillyard, 1984; Federmeier et al. 2007) – with attenuated responses to written or spoken words when they are more predictable from the context. Cloze tasks also clearly engage production mechanisms, as individuals must produce words. For example, words that have a higher cloze probability are initiated earlier in a sentence-final cloze task (Staub et al., 2015), which is commonly taken to indicate that highly active next words are initiated more quickly.

Perhaps unsurprisingly, analysis of cloze probabilities and other word characteristics has led to a lexical focus in much human reading time research, which has largely put to the wayside other dimensions of language processing. Currently, the most common approach to modeling the results of the cloze task has relied on computing the probability of a *specific* word. However, this focus may miss broader features that can explain human reading times and that may influence individuals' word choices. For instance, there is rich semantic structure in the responses that participants provide in the cloze task, such that many preambles support the production of lexically distinct but semantically related words. Even for highly constraining preambles such as (1), participants also produce closely related words in the cloze task (e.g., *gate* and *windows* in addition to *door*). Word-centered approaches necessarily discard this source of variation, largely focusing on which word is the most probable given a particular preamble, rather than the characteristics of the range of possible responses. This variance may be meaningful, as prior research has indicated that incorporating semantic information into models of naturalistic reading can improve predictions of reading times (Luke & Christianson, 2016; Roland et al., 2012). It is conceivable, therefore, that reading the next word following a sentence preamble that produces multiple but closely related continuations may be less challenging than when the types of responses are similarly variable but not related. For example, when reading the sentence "They went to see the famous ____", the word *actor* might be processed slowly due to its low cloze probability, but it may also be because the sentence context produces many possible responses that are not semantically similar to each other (e.g., "painting", "ruins", "accident").

We propose to approach the problem of identifying semantically similar words and clusters of similar words in the cloze task by leveraging recent successes in modeling word meanings using neural networks, specifically large language models (LLMs). Doing so allows us to quantify the frequencies of different linguistic classes in participants' cloze responses and to incorporate those into statistical models of language comprehension (Luke & Christianson, 2016) and production (Staub et al., 2015). In this work, we approximate the semantic properties of each unique response in several cloze datasets by using an LLM to extract vector representations. LLMs extend earlier distributional semantics approaches to approximating word meanings, such as latent semantic analysis (LSA; Landauer & Dumais, 1997) and word2vec (Mikolov et al., 2013), to produce real-valued vector representations that, critically for our purposes, are sensitive to context. Distributional semantics methods allow us to quantify word meanings without specifying semantic features (e.g., that a *door* and *gate* are both "movable barriers"; Rogers & McClelland, 2004; Miller, 1995). Distributional techniques presuppose that two words that typically occur in similar (distributional) contexts are probably similar in meaning and thus have similar vector representations (Brunila & LaViolette, 2022). The similarity in meaning between two words is approximated by some measure of distance (e.g., Euclidean distance or cosine similarity) between the vectors, with more distant vectors being less similar in meaning. Word vectors that are learned by distributional methods have been shown to encode not only aspects of a word's semantics, such as the size of its referent (Grand et al., 2022), but also lexico-syntactic information, such as the word's part-of-speech in context (Rothe & Schütze, 2016), its morphological structure (Mikolov et al., 2013), and overall lexical ambiguity (Beekhuizen, Armstrong, & Stevenson, 2021; Mannering & Jones, 2021).

LLMs present several advantages to modeling the responses that participants provide in the cloze task over approaches like word2vec and LSA. First, LLMs are feed-forward neural networks with many hidden layers. As a result, they process linguistic input in several cascading stages, producing many intermediate vector representations, which differentially encode a wide variety of sources of linguistic variation (Tenney et al., 2019; Rogers et al., 2021). The general view is that the many different representations of a word in different parts or layers of the network are useful for quantifying different aspects of a word's linguistic properties. For example, LLMs encode wordform at lower layers, word sense and syntactic information at intermediate layers, and logical and relational information at the highest layers (Tenney et al., 2019). Given this hierarchical structure, we infer that the later layers of LLMs provide a reasonable approximation of a word's semantics in context. Thus, researchers will find a variety of LLM vector representations for their applications, each indexing different linguistic factors. Second, LLMs produce distinct vector representations of words in every unique linguistic context, which means that the vector representation of the same string (e.g., two instances of the response *door*) will encode subtle variations that echo the other words in the linguistic context (Ethayarajh, 2019). These two advantages allow us to capture participants' different tokens of the "same" response type differently, and also to potentially represent the many different roles that a word can play in the structure and interpretation of a sentence (Jacobs & MacDonald, 2024).

In this work, we collect sentence completion responses in a cloze task, extract the weights from the middle layer from LLM representations of these cloze responses, then apply a clustering algorithm known as a (Bayesian) Gaussian mixture model (BGMM; Roberts et al., 1998) to create several dataset-specific BGMMs. The BGMMs trained on these extracted vector representations of words produce quasi-semantic clusters of responses (e.g., *gate* and *windows* occur in the same cluster). The Gaussian mixture method has been applied broadly in the cognitive sciences to demonstrate the acquisition of different types of linguistic categories (e.g., Li & Joanisse, 2021; Toscano & McMurray, 2010). Our approach to understanding semantic processing in the cloze task relies on an assumption that words that are similar in meaning will occur in similar positions in space and can be clustered on that basis. We use BGMMs to organize the vectors extracted from the cloze responses into clusters in several datasets of cloze responses (Federmeier et al., 2007; Lowder et al., 2017; Luke & Christianson, 2016). That is, we can infer semantic clusters of responses from cloze data (e.g., *door*, *gate*, and *window* may belong to the same cluster) on the basis of how close in space different responses are to each other. Using these clusters, we can obtain an estimate of **semantic** probability, which we define as the probability of a space or set of meanings, including the probability of semantic categories, such as parts of the body, or places one can take a bicycle. Note that this working definition of semantic probability is not intended to refer to logical approaches to semantics, since we are not working with formal definitions of categories (Landauer & Dumais, 1997). Rather, we wish to capture the intuition that words can be both lexically and conceptually predictable. We then incorporate these estimates into regression models of language comprehension and production. We describe RoBERTa and the properties of BGMMs in the next section.

In Experiment 1, we qualitatively and quantitatively demonstrate the coherence of the BGMM's learned clusters across three datasets. Then, we present the results of a coherence rating task to assess the quality of the learned clusters of one of these sets of serial cloze responses (stimuli from Jacobs et al. (2022), derived from Federmeier et al. (2007)). We show that the learned clusters of cloze responses are highly syntactically and semantically coherent, and that participants' ratings of clusters shows that they are particularly coherent at a taxonomic level. Then, in Experiment 2 we show how incorporating additional semantic information beyond word-level predictors improves our ability to explain production data in the cloze task.

Before we present these experiments, we will first explain the modeling framework, from the early stages of sentence processing by the LLM to cluster assignment by the BGMM.

Representing semantic clusters of cloze responses

Our approach extracts word vectors for input to a Gaussian mixture model from the Transformer language model RoBERTa[1] (Liu et al., 2019). While RoBERTa has not been used as extensively in psycholinguistics as models such as GPT-2 (Radford et al., 2019), it has nevertheless proven useful for modeling a wide variety of phenomena, ranging from reading times (Oh, 2021), to neural responses (Mikolov, Coulson, & Bergen, 2022), to linguistic judgments (Mosbach et al., 2020; Lau et al., 2020). We focus on RoBERTa partly because it "encodes" language and can represent how a word fits with the preceding and following context surrounding a particular token. Encoder models may also be interpreted as representing an overall "message" that could be conveyed by the use of a word in a particular context. On the other hand, incremental, next-word prediction models – or decoders – do not predict potential messages; they predict potential

[1] We note that all of the experiments we describe here can be conducted in principle with any variety of LLMs that produce word vector representations, and our experiments are not meant to be an exhaustive exploration of the full space of all language modeling approaches or word vector representations.

signals (e.g., word forms). Thus, from a computational perspective, these models cannot easily assess the impact of a word on a context. The use of RoBERTa has also proven successful in previous related work (for instance, Jacobs & MacDonald, 2023), where RoBERTa was successfully applied to model lexical uncertainty at the message-level. Most pertinently, encoder models (BERT; Devlin et al., 2019) have previously been used to model cloze data, on the basis of the cloze-like objective of the model (Kumar et al., 2021), which we describe below. Our experiments demonstrate the feasibility of using representations from models like RoBERTa for the downstream modeling of language production.

As stated earlier, our goal is to account for the contribution of semantic information to language processing, above and beyond lexical cloze predictability, rather than identifying the model that produces the strongest correlation with human behavior or explains the highest amount of variance in a predictive model sense. For example, we elect to not use the model GPT-2 (Radford et al., 2019), despite its successes in accounting for language comprehension dynamics such as reading times and neural responses (de Varda et al., 2023; Szewczyk & Federmeier, 2022). GPT-2 is strictly serial, processing each token incrementally, and it is primarily optimized to guess how a new word influences the prediction of later words, not to incorporate a word into the broader context of the sentence. That is, when extracting embeddings associated with particular tokens, the embeddings at the highest levels encode GPT-2's predictions about next words. Using the embedding representation of the final word of a sentence from GPT-2 is therefore oriented toward next potential tokens, and not the current one. For this reason, we opt to not use models like this in the present work. Impressionistically, the mechanisms that generate representations in RoBERTa are more closely aligned with our modeling goal. However, we note that the clustering approach we take here can in principle be applied to any model that can produce a representation of cloze responses.

Extending earlier approaches such as word2vec (Mikolov et al., 2013), RoBERTa's architecture is a multi-layer, feed-forward neural network whose primary objective is to predict which word fits best in a given linguistic context. RoBERTa (Liu et al., 2019) is an encoder language model that uses the Transformer architecture, which uses a mechanism known as attention to capture pairwise interactions between words or tokens (Bahdanau, Cho, & Bengio, 2014; Vaswani et al., 2017), and a masked language modeling objective to predict those tokens. The attention mechanism critically allows models to capture pairwise interactions between the words in the input, and thus build a higher-order representation of a sentence's structure (Merrill et al., 2020; Weiss et al., 2021). This simple masked word prediction training mechanism allows for the powerful learning of paradigmatically similar representations of words in context (Baroni, Dinu, & Kruszewski, 2014).

In all of our experiments, RoBERTa takes each sentence as input; the sentence is broken up into its component subwords, whose representations cascade forward, incrementally taking into account contextual information such as a word's part of speech. In our experiments, we identify the embeddings for the tokens associated with specific cloze responses, focusing on the 7th layer of the network, as this layer is highly sensitive to linguistic structure of all kinds (Chronis & Erk, 2020; Miaschi et al., 2020). The token vectors that RoBERTa produces have 768 real-valued dimensions. In order to capture the whole-word properties of each cloze response, such as its length and morphological structure, we average the embeddings across all subword tokens for a given cloze response. For example, if RoBERTa tokenizes a cloze response such as *pizzas* into the subwords sw_1 = *pizz* and sw_2 = *-as*, then we take the mean of each dimension across both vectors. This subword averaging approach has been previously used (Ács et al., 2021) and produced acceptable results for the clustering task in early development of the work.

Finally, we provide all of the unique embeddings for a given set of cloze responses as features to train a BGMM, as diverse embeddings from multiple contexts help the model to estimate clusters of cloze responses across the full range of response profiles in each cloze task. At a high level, a Gaussian mixture model attempts to identify all of the data-generating distributions that gave rise to the vector representations of the responses observed for a given cloze experiment (McLachlan & Basford, 1988). That is, the goal of this clustering algorithm is to identify the types of responses that participants can choose to make, which depends in part on the linguistic context, as well as the specific lexico-semantics of the words being produced. We note that we do not, and cannot, bias the model with repeated responses for words that many participants guess (i.e., give more weight to words with higher cloze probability). Rather, the model treats each response as equal regardless of its cloze probability due to the fact that the model computes the similarity (covariance matrix) between all pairs of data points. If any data points are identical, as in repeat responses, the model will find perfect correlations and cannot converge. In any case, our goal is to model the characteristics of the semantics of the data, regardless of how predictable the individual responses are.

Incorporating LLM representations into a BGMM presents several major advances over prior work. First, Gaussian mixture models, unlike topic models, are well-suited to continuous vector spaces such as word embeddings (Sridhar, 2015), which represents a major advance from more symbolic, abstractionist earlier approaches. For instance, the Latent Dirichlet Allocation (LDA) algorithm has been applied to retrieval-oriented cognitive tasks, as in recognition memory experiments (Steyvers, Griffiths, & Dennis, 2006; Griffiths, Steyvers, & Tenenbaum, 2007). This approach, often referred to as topic modeling, operates over discrete entities (words tokenized from text) and generally requires more data points than a standard cloze dataset to produce coherent "topics", analogous to clusters. However, the BGMM presents an advance over topic modeling that integrates both the Dirichlet mechanism and extends easily to continuous vector spaces, such as word and sentence embeddings (Rasmussen, 1999; Roberts et al., 1998). The Dirichlet prior makes it possible to infer clusters in a dataset of many different sizes, ranging from clusters with no data points to those with many, depending on the configuration. This flexibility is important for modeling cloze responses, as some sentence contexts may be very semantically constrained whereas others may be broad. Finally, many LLMs, in contrast to earlier word-based neural language models (e.g., Goodkind & Bicknell, 2018), rely on subwords with pre-trained weights, which enables us to generate embeddings for nearly any input. In fact, in the current work, cloze responses containing more than one word, proper names, misspellings, or non-normative spellings (e.g., *i* instead of *I*) were often combined into a distinct cluster from responses that fit more closely with the context or that did not contain misspellings. Depending on the researcher's needs, cleaning cloze response data may be necessary to avoid producing these clusters; however, such a cluster may be interesting to psycholinguists in its own right.

It is important to understand how Gaussian mixture models function to fully explain the motivation for the clustering approach used here.[2] Put simply, Gaussian mixture models are an unsupervised statistical technique that estimate combinations, or mixtures, of Gaussian distributions on the basis of numerical data (McLachlan & Basford, 1988). The Gaussian mixture

[2] We refer the reader to resources such as Reynolds (2009) – Reynolds, D. A. (2009). Gaussian mixture models. Encyclopedia of biometrics, 741(659-663) – and many other textbook resources (e.g., Bishop, 2006, *Bishop, C.M. (2006). Pattern Recognition and Machine Learning (Information Science and Statistics).*).

model specifically estimates hypothesized means, variances, and covariances that are generated by some number of Gaussian distributions that maximize the likelihood of the data on the basis of the densities of datapoints. For example, consider the distributions shown in Figure 1, created by combining Gaussians with different means and variances. The goal of a Gaussian mixture model would be to identify these underlying Gaussian distributions (their means and variances) that were *mixed* to create the distributions of data, here rendered as the density of observation values.

(a) (b) (c)

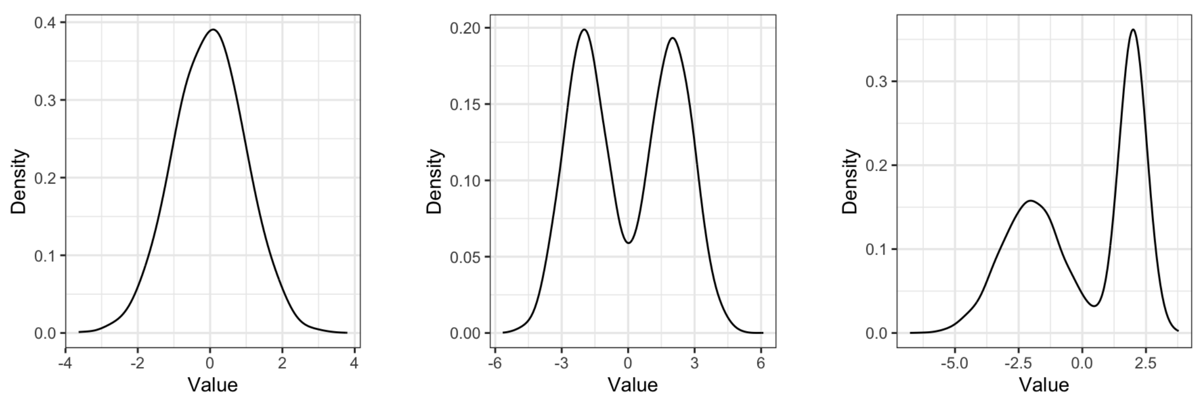

Figure 1. Density plots of Gaussian-distributed data with different numbers of Gaussians of varying means and variances. a) A simple Gaussian with $\mu = 0$ and standard deviation $\sigma = 1$. b) Two combined Gaussians with different means ($\mu_1 = -0.5$, $\mu_2 = 0.5$), but identical variances ($\sigma_1 = \sigma_2 = 1$). c) Two combined Gaussians with different means ($\mu_1 = -2$, $\mu_2 = 2$) and different variances ($\sigma_1 = 1.25$ and σ_2=0.5).

Gaussian distributions are not restricted to the univariate domain; multivariate Gaussians are also possible, with each of the *n* dimensions of the Gaussian having its own mean and standard deviation, in addition to a variance-covariance matrix that explains the nature of the correlation between each dimension and all the others (see Figure 2). The objective of the Gaussian mixture model for multivariate data remains the same: to recover the (Gaussian) distribution(s) that generated the data. Here, the data are assumed to be composed of a *mixture* of Gaussian processes, or distributions. Thus, for multivariate data as in Figure 2, our aim would be to recover the mean positions and slopes of both clusters in both x and y dimensions, which is possible with the Gaussian mixture model. This is necessary for our purposes, as most word embeddings have a very high dimensionality, ranging from 100 on the low end to several thousand for the largest models. As noted above, our experiments construct Gaussian mixture models over the 768 dimensions that result from extracting embeddings from the base RoBERTa model (Liu et al., 2019).

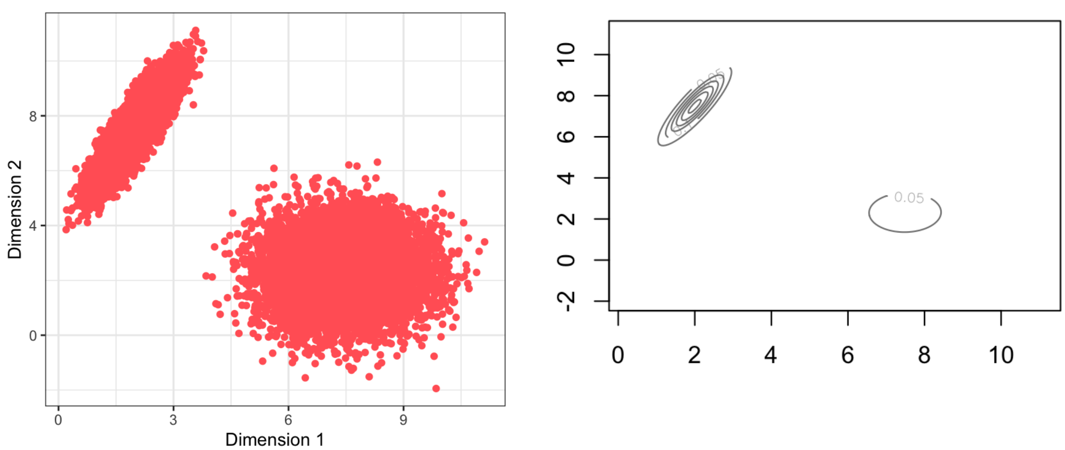

Figure 2. Two idealized multivariate Gaussian distributions plotted in two dimensions. Left panel represents raw synthesized data over 20k observations, half in each of the above clusters. Right panel represents recovered density parameters from the R package `mclust` (Fraley et al., 2024; version 6.1) and the function `densityMClust` to recover the Gaussian distributions. The model correctly infers two equiprobable

categories with means (7.49, 2.31) for the first cluster, and (1.99, 7.49) for the second cluster.

Most off-the-shelf implementations of GMMs require specifying the number of clusters to estimate prior to clustering. However, there is fundamental uncertainty about how many data-generating processes (mixtures) there are in a given dataset that could explain patterns of cloze responses. Given the skewed distribution of human responses in the cloze task, with some sentences or contexts being highly constraining and others not, it is critical to have a model that can flexibly and robustly assign words to clusters that may vary in size, to the extent that doing so increases the fit of the model to the dataset of cloze responses. Furthermore, cloze data follow a multinomial distribution, such that every word is assigned some probability (potentially 0) of being produced in a specific linguistic context. Thus, we opt to use a Bayesian implementation of Gaussian mixture models (BGMMs), which allows us to introduce additional priors that allow the model to select as many clusters as it needs and allocate as many data points as needed to those clusters. Specifically, by introducing a Dirichlet prior, the conjugate prior of the multinomial distribution, BGMMs can be configured to cluster more or less aggressively, lumping as many data points into as few clusters as possible, or producing more uniform distributions, with most clusters being of equal size. Selecting a good value for this weight concentration prior is critical, as it determines how densely each cluster is filled. We use the same weight concentration prior for each cloze dataset, varying only the number of clusters learned on the basis of qualitative inspection of cluster coherence, which appears to vary primarily based on whether the stimuli are artificially constructed single sentences (Federmeier et al., 2007) or more naturalistic texts composed of multiple sentences (Luke & Christianson, 2016, 2018; Lowder et al., 2017). Supplemental materials on OSF detail the specific implementation of our BGMMs at https://osf.io/n4avt/.

In the experiments that follow, we first demonstrate the feasibility of this clustering approach and present quantitative evidence that the clusters that the BGMM learns are coherent and reflect relevant sources of variation in the cloze task. Then, we move to analyses that incorporate cluster-level properties in predicting production times in the cloze task, and show that the addition of cluster information helps to account for semantic competition and facilitation in cloze production. The results reveal that semantic fit – independent of lexical fit – influences processing fluency and demonstrate the utility of the method.

Experiment 1A

Human responses in cloze experiments vary considerably across different variants of the task, particularly when the provided context is not constraining. Numerous prior studies have obtained cloze probability estimates to approximate the predictability of a word in context, either using a continuous, serial ("cumulative") cloze task, in which participants guess each word in order in a sentence (Luke & Christianson, 2016, 2018; Lowder et al., 2017), or using a stem completion task in which participants are provided with a sentence fragment and are asked to guess the identity of the final word (Federmeier et al., 2007).

Since the creation of the cloze task (Taylor, 1953), there has been an extensive literature comparing different types of cloze tasks and examining the correlation between the resultant cloze probabilities with sentence processing difficulty (Taylor, 1953; Tuinman, Blanton, & Gray, 1975). However, and more importantly for our research question, it is unclear how easily we can apply a singular procedure for clustering cloze responses across datasets. For example, existing datasets of cloze responses not only vary in the tasks used to obtain the responses, but the types of stimuli and participant pools, which could limit the ease of constructing a BGMM with general-purpose guidelines. It may be the case that presenting the cloze task in a different way or with different types of stimuli leads to very different responses from the participants completing the task. If the BGMMs trained over word embeddings can only produce coherent clusters for datasets that contain a certain degree of structure and variation, this limits the utility of our method. To assess this, here we use three cloze datasets varying in their structure and complexity and examine clustering of cloze responses using BGMMs trained on LLM embeddings.

Method

In order to demonstrate the feasibility of clustering participant responses for data obtained from different cloze tasks, we train separate BGMMs for three serial cloze datasets. Our BGMMs were implemented in scikit-learn (version 1.3.2) and embeddings were extracted using the huggingface transformers package (version 4.36.2) from the RoBERTa base model. In all cases, the model results we report hold the weight concentration prior constant, and the final models only vary in the cloze data that they are trained on and the number of Gaussians that are available to the model to assign participant responses. Our selection of weight concentration prior was informed by informally inspecting clustering behavior while fixing the number of clusters on the Jacobs, Hubbard, and Federmeier (2022) cloze dataset. A value of 0.1 was fixed for all subsequent models and datasets. The exact number of clusters for each dataset was then determined qualitatively for each dataset by visually inspecting the content of the learned clusters, both for preambles in each dataset as well as the broader composition of clusters across different linguistic contexts.

First, we examine serial cloze responses based on procedures described in Jacobs et al. (2022) to the 282 “Cloze Constraint (CLOC)” sentences from Federmeier et al. (2007). The

CLOC stimuli were designed by experimenters and varied in the predictability of the sentence-final word, with sentences leading to lower-probability final words (< 42% cloze probability) designated *Weak Constraint* and those leading to high-probability completions (>67% cloze probability) as *Strong Constraint*. We additionally apply our clustering method to serial cloze datasets that are more representative of real-world texts. We cluster serial cloze responses from Lowder et al. (2017), who selected 40 multi-sentence texts used in reading level assessments, as well as serial cloze data from the Provo Corpus (Luke and Christianson, 2016, 2018), which comprised 55 multi-sentence texts taken from news sources. In all datasets, each cloze context or prompt for a response has an average of 40 participants per text, although datasets range in size and lexical diversity, from the low thousands of unique context-response pairs (Federmeier et al., 2007) to tens of thousands (Lowder et al., 2017; Luke & Christianson, 2016, 2018), depending in part on the length and number of sentences per text. We include representative stimuli from these studies in Table 1 below.

Table 1. Example sentences from serial cloze task corpora. Note that in the original Federmeier et al. (2007) stimuli, *drought* was the most common cloze response for this preamble. However, in the Jacobs et al. (2022) data, *fire* was the most probable response.

Stimulus set	Example sentence
Jacobs et al. (2022); Federmeier et al. (2007)	"Most of the trees died after last year's fire."
Luke and Christianson (2016)	"There often seems to be more diving in soccer than in the Summer Olympics."

Lowder et al., (2017)	"Gargantuan housing developments sprang up on which quite a fantastic number of families were accommodated."

Results

Initial clustering results from the three datasets appear promising at first glance. We present examples of the clusters learned by the model in Table 2. The reader will note several qualitative patterns that can be observed that are consistent with the training objective of prediction-based LLMs (Baroni, Dinu, & Kruszewski, 2014), such as the tendency for words in a cluster to share morphological properties (e.g., plurality) or grammatical category (e.g., nouns). Indeed, a wide variety of neural language model interpretability studies have shown that part-of-speech information is encoded by the hidden layers in models like BERT and RoBERTa (Tenney et al., 2019). The finding that many of the clusters learned over the Federmeier et al. (2007) stimuli are comprised of a dominant lexical category is therefore unsurprising.

Table 2. Words sampled from BGMM clusters from several cloze datasets.

Federmeier et al. (2007)	Luke and Christianson (2016)	Lowder et al. (2017)
fire	solids	dog
drought	shells	horse
storm	neutron	chicken
hurricane	shell	bear
harvest	receptors	cat
flood	gas	fish
fires	impulses	mouse
frost	signals	penguin
	gases	pigeon
	carbon	tiger

Despite good qualitative results, a closer inspection of the linguistic and structural properties that characterize the contents of each cluster is necessary, which we quantify using several different measures. In Table 3, we present estimates of the number of word types per cluster; number of clusters per preamble; number of words per cluster per preamble; and part-of-speech diversity within each cluster. Part-of-speech diversity was determined by computing the proportion of the tags for each response that belonged to the most frequent part-of-speech; we approximated this using the POS-tagged SUBTLEX-US dataset (Brysbaert, New, & Keuleers, 2012) and extracted each string's most frequent POS tag. We present two distinct clusterings of the Jacobs et al. (2022) response data, derived from the Federmeier et al. (2007) stimuli. The first clustering (k=100) represents this study's first attempt to cluster the data, which we term Federmeier-100. We also present Federmeier-282, which was designed to capture the idea that each complete sentence fragment (n=282) from the original CLOC stimuli might contribute uniquely to the clustering behavior of the model. Clusterings of the same stimulus set with different numbers of clusters reveals that higher numbers of clusters influence the distribution of words in clusters by allocating fewer words to each cluster. All analyses presented in later experiments rely on Federmeier-282. Importantly, the clusters contain high degrees of coherence at the lexico-syntactic level, with clusters generally favoring a single part of speech.

Taken together, the results described in Table 3 provide convincing evidence that clustering is taking place both at the level of the entire dataset, with clusters generally containing multiple words, as well as within sentences, such that many preambles contain clusters with multiple words. We provide visualization of these distributions from the Federmeier et al. (2007) stimuli and serial cloze responses from Jacobs et al. (2022) in Figure 3. We include visualizations of these statistics for Luke and Christianson (2016, 2018) and Lowder et al. (2017) in the Supplementary Materials.

Table 3. Descriptive statistics characterizing linguistic properties of three different cloze datasets. Clusterings of the same stimulus set with different numbers of clusters (Federmeier-100 vs. Federmeier-282) reveal that higher numbers of clusters influence the distribution of words in clusters.

Stimuli	# of clusters	Median # of word types per cluster	Mean # of clusters per preamble	Mean # of words per cluster per preamble	Part-of-speech diversity (Proportion most frequent POS)
Federmeier et al. (2007)	100	9.5	8.01 +/- 4.29	1.60 +/- 0.82	.89 +/- .17
Federmeier et al. (2007)	282	6	9.91 +/- 6.00	1.26 +/- 0.43	.92 +/- .14
Luke and Christianson (2016)	125	18	9.01 +/- 3.87	1.74 +/- 1.00	.84 +/- .21
Lowder et al. (2017)	250	11	9.15 +/- 4.72	1.59 +/- 0.89	.86 +/- .18

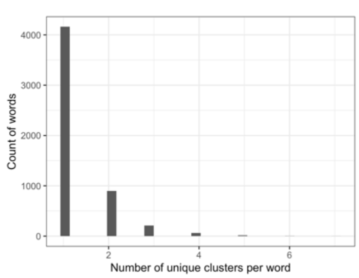

Figure 3A. Distribution of number of unique clusters for a given word string for Federmeier-282.

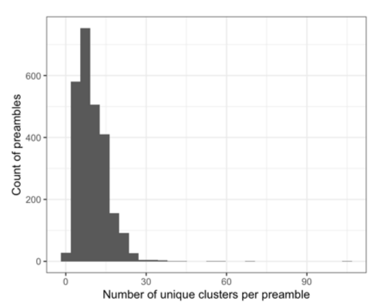

Figure 3B. Distribution of number of unique clusters per preamble for Federmeier-282.

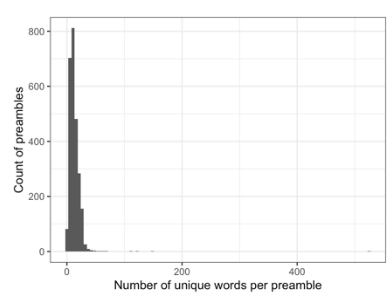

Figure 3C. Distribution of number of words given preamble for Federmeier-282.

Due to the nature of the serial cloze task, part of speech diversity across an entire dataset is quite high, and it is important to show that the tendency to favor a single part-of-speech is meaningful. Below we visualize the average part-of-speech diversity of the clusters and the part-of-speech diversity of the full dataset by computing entropy over the proportion of part-of-speech tags in the cluster or the overall dataset (i.e., -$\sum p\ log(p)$). The part-of-speech entropy of the whole dataset is 0.734, while the majority of all clusters (150/282) have no variance (i.e., 0) in part-of-speech at all. In Figure 4 below, the overwhelming majority of clusters in the Federmeier-282 data set have lower entropy than the global distribution, demonstrating that the embeddings that we use capture lexico-syntactic properties of the words.

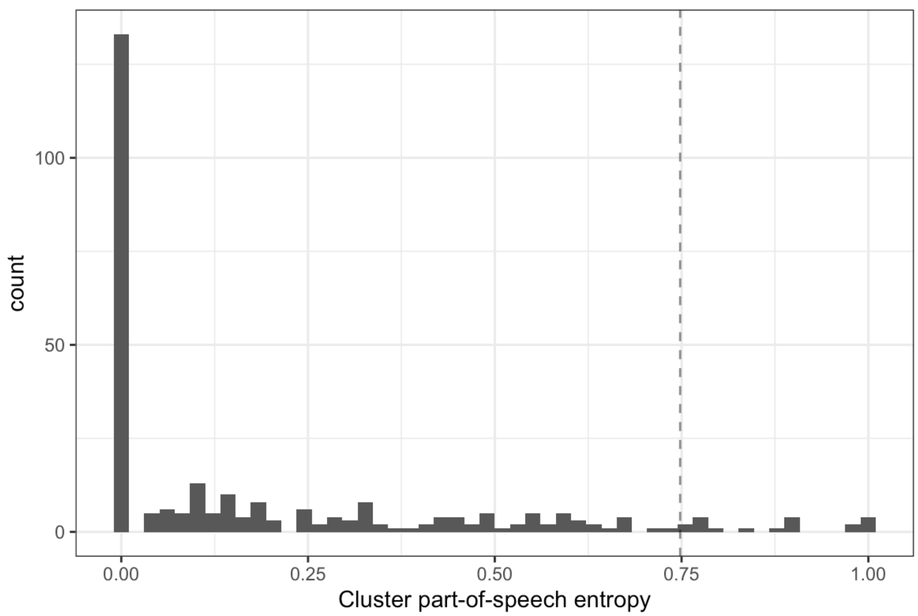

Figure 4. Distribution of part-of-speech entropies for serial cloze clusters from Federmeier et al. (2007) stimuli. Dashed line represents dataset-wide entropy.

Discussion

Experiment 1A provided promising results that the clusters derived from the Bayesian Gaussian Mixture modeling approach are coherent, even across different serial cloze datasets comprised of highly controlled, naturalistic, and real-world texts. Namely, even though the BGMMs that were trained on the three different datasets have different numbers of clusters, the parameters that constrain the diversity and prevalence of clusters produce qualitatively similar clustering behavior across the three datasets with identical weight concentration priors, demonstrating that the method is broadly applicable to serial cloze tasks using different types of stimuli. It may be of value to researchers to determine the optimal number of clusters empirically; we return to this later again in the Discussion. We see this as promising evidence that the BGMM approach can provide a reasonable basis on which to model semantic processing in language comprehension and production, as in Experiment 2.

The evaluation of Experiment 1A was fairly informal, and judging the clusters simply by POS diversity may not be sufficient to determine whether the clusters are semantically coherent. Thus, we evaluate the clusters' coherence using human ratings in Experiment 1B in order to validate the quality of the clusters above and ensure that the clusters are more human-interpretable than random sets of words. Experiment 1B also assesses the extent to which the clusters show taxonomic and/or thematic similarity, as these different dimensions of semantic coherence may meaningfully influence language processing.

Experiment 1B

Informal evaluations of the quantitative properties of the learned clusters from the BGMMs from Experiment 1A provided some insight into the structure of the responses that participants provide. However, a major concern for the application of our method is that the clusters themselves should be semantically coherent in a human-relevant way. Qualitatively, the clusters produced by the BGMMs appeared semantically coherent (see Table 2); however, it is critical to empirically measure the level and type of coherence in the clusters using participants who are naïve to the source and practical application of this coherence. Here we wished to determine that the clusters learned by the BGMM are coherent in a practical and measurable way, not purely guided by experimenter intuition or qualitative assessment. In practice, the method could have clustered words that did not differ in their coherence compared to random “foil” clusters, and so Experiment 1B provides a quantitative assessment of the qualitative results of 1A.

In Experiment 1B, we examine semantic coherence empirically, using a clustering of the serial cloze data from Jacobs et al. (2022), which is derived from the stimuli in Federmeier et al. (2007) described above, and for which we have production times in Experiment 2. The CLOC stimuli were explicitly constructed for psycholinguistic experiments studying lexical prediction in sentence comprehension and have a tighter degree of control, in contrast to the Luke and Christianson (2016, 2018) and Lowder et al. (2017) data sets, which are more naturalistic, and lack timing data. Furthermore, the CLOC stimuli have been used in numerous studies on prediction and semantic processing in online language comprehension tasks, including not only those measuring event-related potentials (ERPs) with rapid serial visual presentation methods but also in the context of self-paced reading (Payne & Federmeier, 2017) and with varied participant populations, including older adults (Wlotko & Federmeier, 2012); thus, these stimuli not only have been experimentally validated, but also the clusters derived from these stimuli can be related to psycholinguistic measures. Additionally, since the weight concentration prior of the BGMM was set to be the same across experiments, it is unlikely that the cluster coherence would vary wildly across different sets of stimuli, and thus an evaluation of the CLOC stimuli is likely to be representative of the effectiveness of the method.

In this experiment, we recruited a different set of online participants to complete a semantic coherence judgment task and asked them to rate the overall semantic coherence of the clusters. We also asked them to more specifically assess the degree of taxonomic and thematic similarity in the learned clusters. By defining taxonomic and thematic similarity for the participants and inducing them to think about those dimensions in particular, we believed we could help ground the coherence rating task for the participants and increase consistency in what participants are attending to. Moreover, the more specific ratings help us not only to ensure that the clusters derived from the BGMM were sensible and semantically coherent, but also to determine whether the BGMM method learns clusters based on taxonomic and/or thematic similarity, which may differentially influence language comprehension and production.

Taxonomic similarity is one dimension of semantic variability that is relevant to human language processing (Mirman, Landrigan, & Britt, 2017). Words that share some taxonomic features (e.g., both *cat* and *dog* are kinds of pets) are assumed to have similar semantics relative to words that do not (e.g., *cat* and *refrigerator*). Moreover, this dimension of similarity can be conceptualized as *paradigmatic*, in that words within the same taxonomic category may often be substituted for each other (e.g., "Their cat/dog/mouse is so cute."). It is well-known that word embeddings are highly sensitive to paradigmatic similarity, as the training objectives of most neural network methods rely on (indirectly) predicting which words are similar to each other in a particular linguistic context (Asr, Zinkov, & Jones, 2018). Thematic similarity, on the other hand, captures the idea that words can be similar by occurring together; cats and witches are not generally substitutable for each other, but in contemporary American culture often are seen together around Halloween time. Similarly, the word *pumpkin* may be thematically related to *cat* and *witch*, though it shares few properties with these other words. Thus, thematic and taxonomic similarity reflect different aspects of the relationships between words and can impact language processing differently.

Our goal for the coherence judgment task was to not only determine the overall coherence of the clusters derived from the BGMM, ensuring that they are not random or poorly constructed, but also to identify the degree of taxonomic vs. thematic similarity of the words in each cluster. Importantly, demonstrating that the clusters created by LLMs have a high degree of taxonomic similarity would provide support for the use of these clusters in models that incorporate semantic factors that influence language processing, as taxonomic similarity can be considered to reflect overlap of semantic features of words.

For the rating study in Experiment 1B, we constructed "coherent" and "foil" clusters out of the Federmeier-100 clusters learned over the CLOC serial cloze responses (Jacobs et al., 2022; Federmeier et al., 2007). Foil clusters were clusters of words derived by sampling from the coherent clusters, created with the purpose of comparison to the clusters derived by the BGMM. Our approach is similar to a permutation testing approach – if the clusters formed by the BGMMs are generally incoherent, then participants will give similar coherence ratings to the actual clusters and the foil clusters. We expected to observe greater similarity ratings for coherent clusters compared to foil clusters, and we also expected to see a positive relationship between ratings of overall cluster coherence and taxonomic similarity. Thematic similarity may be less successfully captured by LLMs and may be influenced by the greater degree of variability of the kinds of words that would appear in taxonomically coherent clusters; therefore, our expectations regarding thematic similarity ratings were less clear. For instance, a foil cluster could unexpectedly have high thematic similarity, as the words randomly selected for the cluster could unintentionally be related in some way. That being said, we overall expected greater ratings of thematic similarity for coherent clusters compared to foil clusters.

Method

Stimuli

We selected 58 of 100 "coherent" clusters for inclusion as stimuli if the dominant part-of-speech (POS) of the words within the cluster was either an adjective, noun, verb, or proper name, as determined by the SUBTLEX-US norms (Brysbaert, New, & Keuleers, 2012). Additionally, clusters were constrained to include only a single POS category, or, in the case of multiple POS categories, if the most frequent POS was at least twice as frequent as the next-most frequent POS category. These inclusion criteria produced 58 "coherent" clusters for evaluation. We then assigned coordinates to each word in the cluster using two random samples from a normal distribution for an arbitrary "x" and "y" coordinate in order to visualize the cluster for presentation during the experiment. This created clusters such as Figure 5A.

For the creation of "foil" clusters, we randomly sampled one word from each of the available clusters until the cluster reached the same number of words as the "coherent" cluster. We used the same rendering procedure to assign random coordinates to each word, as in Figure 5B.

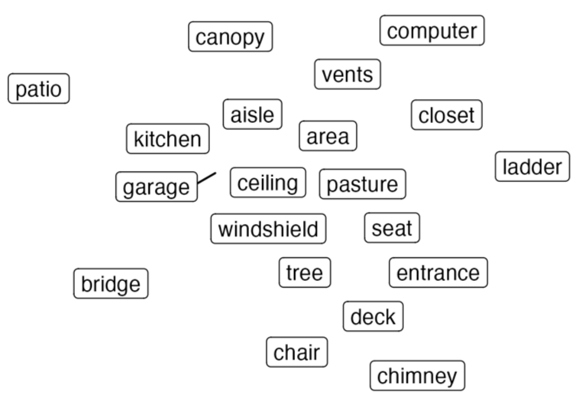

Figure 5A. Coherent cluster example.

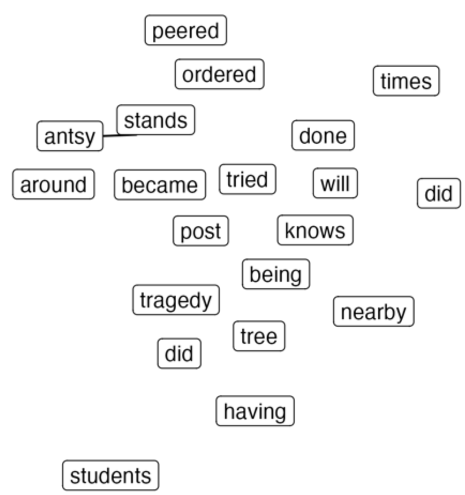

Figure 5B. Foil cluster example.

Participants

We recruited 62 participants from Prolific who self-identified as having learned English as their first language and lived in the U.S. Participants received $5 USD in compensation for completing a 20-minute task. All participants provided informed consent approved by the University of Illinois at Urbana-Champaign Institutional Review Board.

Procedure

Upon registration in the study, participants were forwarded to a Qualtrics survey page at which they provided informed consent. From there, they were given instructions on the rating task, which included example clusters that demonstrated the difference between taxonomic and thematic similarity. We provide the instructions that participants received in Appendix A, which included clear descriptions of taxonomic and thematic "dimensions" of semantic similarity. Following the instructions, participants rated the semantic coherence, taxonomic similarity, and thematic similarity of 58 clusters that were presented as images. Each cluster was randomly selected as either the Coherent (BGMM-generated) cluster or its randomly composed Foil (see Figure 5). Participants rated each of the three questions on a 1-5 Likert scale. For each trial of the experiment, participants were presented with a Coherent or Foil cluster on the screen and were prompted to answer the question, "The words in this cluster go together" with values from "Not well at all" (1) to "Extremely well" (5). For the taxonomic and thematic similarity rating tasks, participants answered the question, "Do the words in this cluster show either of the below special dimensions of similarity?" and were shown a scale for both taxonomic and thematic similarity ranging from "None at all" (1) to "A great deal" (5). Upon completion of the study, they received a short debriefing explaining the purpose of the study.

Results

We present two analyses of the ratings data. First, we compared Coherent and Foil clusters for their overall cohesiveness (the first rating participants gave). Then, we tested whether participants discerned greater taxonomic and/or thematic similarity in Coherent relative to Foil clusters, while taking into account their rating of the overall coherence of a particular cluster. We conducted cumulative link mixed model analyses using the ordinal package (Christensen, 2023; version 2023.12-4) in R (version 4.3.1) with all categorical factors centered using difference coding to facilitate model convergence. All analyses that we present here contain maximal random intercepts and slopes by participant and intercepts by cluster.

Participants rated Coherent clusters as significantly more cohesive than Foil clusters, typically rating Coherent clusters in the 3-5 range whereas Foil clusters were in the 1-3 range. However, some clusters that the BGMM learned for the CLOC responses were rated as having low coherence; further investigation is needed to determine why some BGMM clusters produce lower-quality clusters, or what led participants to provide these ratings. We present the broad distributions of ratings in Figure 6 and Coherent clusters that were rated as having high- and low-coherence in Figure 7. The cumulative link model confirmed this general pattern (Table 4).

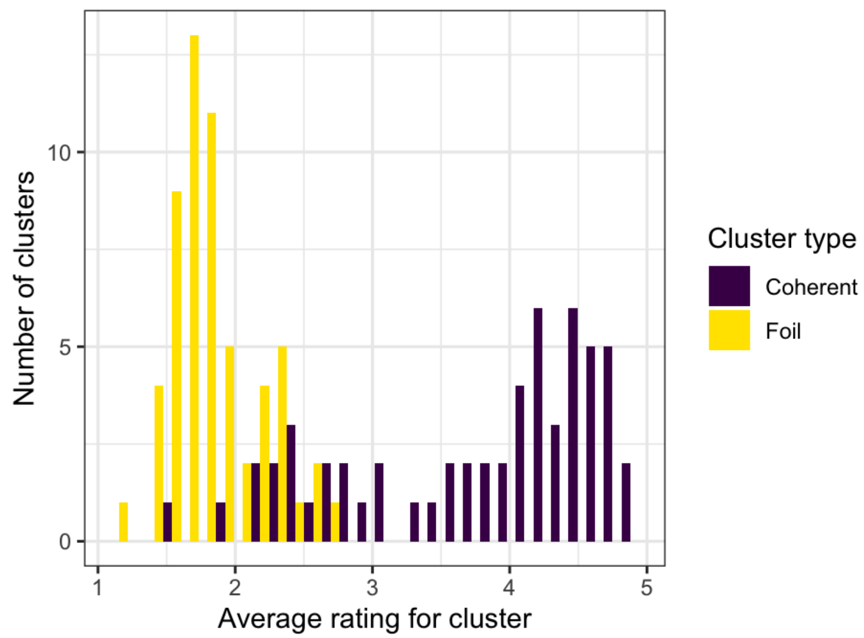

Figure 6. Cluster coherence rating histogram for coherent and incoherent clusters.

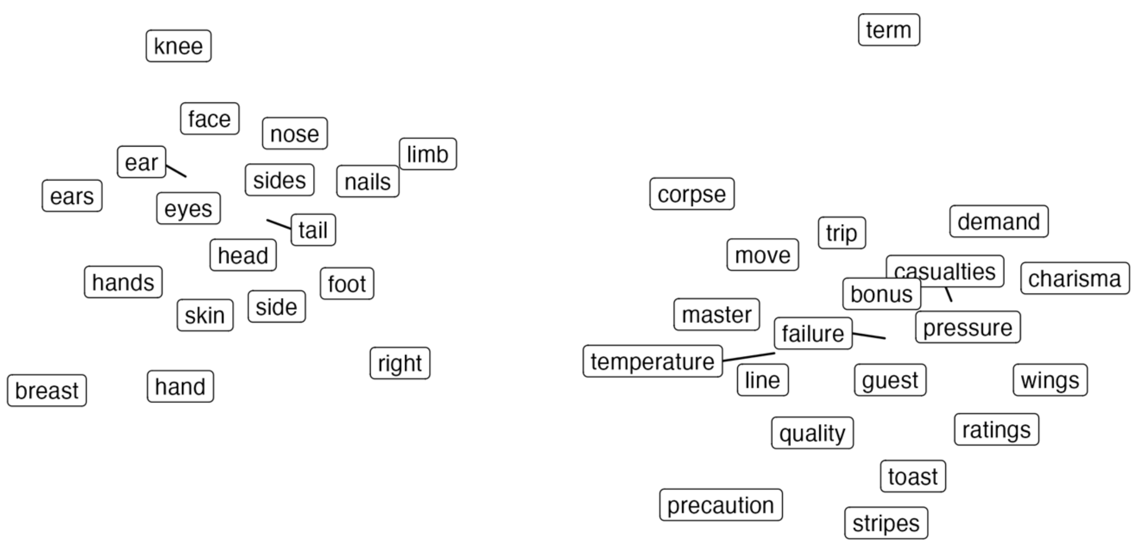

Figure 7. Example of a high-coherence Coherent cluster (left) and low-coherence Foil cluster (right).

Table 4. Cumulative link mixed effects model of cluster coherence.

Formula: Coherence Rating ~ (1 + Trial Type | Participant ID) + (1 | Cluster) + Trial Type

Fixed effect	Contrast	Estimate	SE	*z*	*p*
Trial Type (Coherent vs. Foil)	Difference	4.00	0.26	15.6	< .001
	Threshold coefficients	Estimate	SE	*z*	*p*
	1 | 2	-1.87	0.20	-9.50	< .001
	2 | 3	-0.09	0.19	-0.46	n.s.
	3 | 4	1.30	0.19	6.67	< .001
	4 | 5	2.83	0.20	14.08	< .001

In addition to the analyses of coherence ratings showing that BGMM-trained clusters have greater coherence than Foil clusters, we also assessed the taxonomic and thematic similarity of Coherent versus Foil clusters as a function of cluster type and the original coherence rating. Overall, coherence ratings strongly correlated with the degree of taxonomic similarity identified by participants; thematic similarity demonstrated a less straightforward relationship to coherence. Moreover, Foil clusters had lower taxonomic similarity at all levels of coherence, while this pattern was also more complicated for thematic similarity. At higher levels of coherence, visual inspection suggests that Foil clusters appeared to demonstrate higher thematic similarity than Coherent clusters; this may be due to the random selection procedure for generating foils. We visualize these effects in Figure 8.

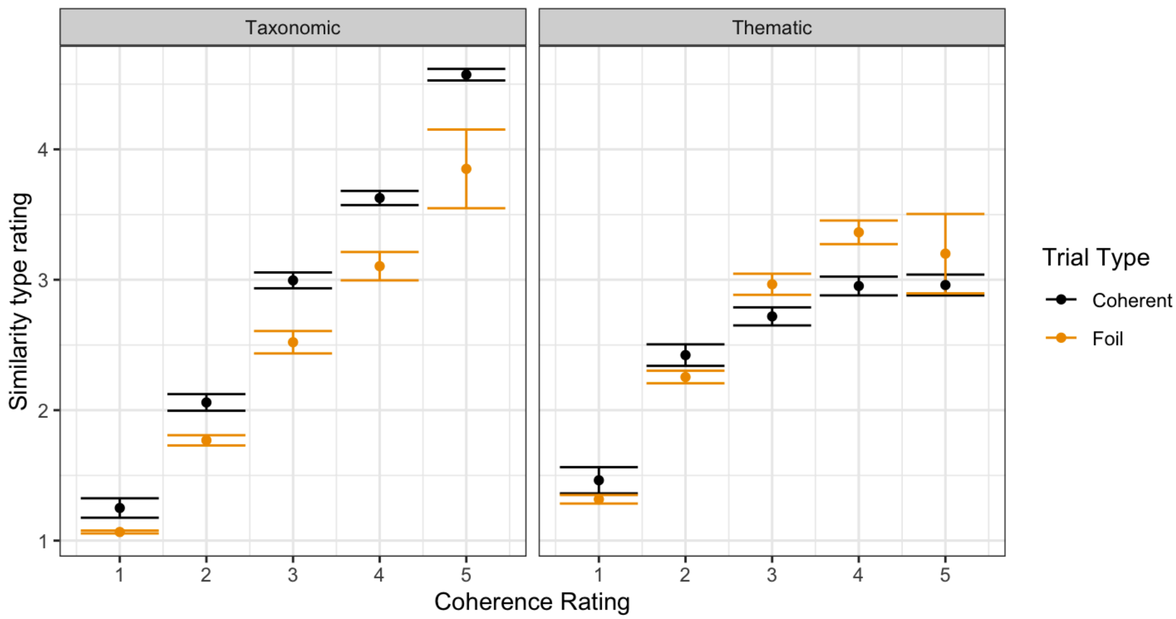

Figure 8. Relationship between coherence rating and taxonomic and thematic similarity ratings for coherent clusters and their foils.

A cumulative link mixed model of the rating data for taxonomic and thematic similarity revealed significant main effects and interactions for all factors; we present these results in Table 5. Coherent clusters were rated as more similar than Foil clusters, and the more coherent a cluster was, the higher the similarity score tended to be. Taxonomic similarity ratings were generally higher than thematic similarity ratings, and the effect of coherence by trial type was non-trivial. Surprisingly, there was a significant three-way interaction between a participant's coherence rating of a cluster, whether they were rating taxonomic or thematic similarity, and whether the cluster was a foil or BGMM-generated. Overall, the results suggest that the clusters broadly capture taxonomic similarity, such that more coherent clusters typically demonstrate featural similarity-based (family-like) semantic relationships, rather than situational or event-based similarity.

Table 5. Cumulative link mixed effects model of cluster coherence.

Formula: Rating ~ (1 + Coherence Rating * Trial Type * Similarity Type | Participant ID) + (1 | Cluster) + Coherence Rating * Trial Type * Similarity Type

Fixed effect	Contrast	Estimate	SE	*z*	*p*
Coherence rating	Numeric	1.62	0.10	15.44	< .001
Trial Type (Coherent vs. Foil)	Difference	1.86	0.26	7.22	< .001
Similarity Type (Thematic vs. Taxonomic)	Difference	2.16	0.44	4.94	< .001
Coherence * Trial Type		-0.62	0.11	-5.53	< .001
Coherence * Similarity Type		-0.65	0.16	-3.96	< .001
Trial Type * Similarity Type		2.15	0.55	3.85	< .001
Coherence * Trial Type * Similarity Type		-1.42	0.24	-5.73	< .001
	Threshold coefficients	Estimate	SE	*z*	*p*
	1 | 2	2.42	0.32	7.48	< .001
	2 | 3	4.67	0.33	14.13	< .001
	3 | 4	6.38	0.33	18.88	< .001
	4 | 5	8.55	0.35	24.33	< .001

Discussion

Taken together, the results of Experiment 1B demonstrate that the BGMM clusters have a high degree of coherence, and that this coherence is predominantly driven by taxonomic similarity. This relationship may be due to the fact that clustering was performed on word embeddings from an LLM, which may be more sensitive to taxonomic relationships between words compared to thematic relationships. More importantly, the high degree of coherence of the learned clusters provides a validation of the qualitative results of Experiment 1A, which evaluated the clusters in an informal way.

Although our main goal was to assess the feasibility of using the BGMM to cluster vector representations of cloze responses in ways that align with human semantic processing, the relative weight of taxonomic versus thematic similarity may nevertheless also be relevant. For instance, the Dell (1986) model of speech production relies on both *paradigmatic* and *syntagmatic* similarity to capture patterns of speech errors. Similarly, computational models of language production (e.g., Oppenheim, 2023) that account for semantic interference use semantic features that are shared within categories (e.g., animals). While it is clear that the BGMM's performance needs to be sufficiently high, it is unclear if the level or type of coherence of a cluster meaningfully affects production times in the serial cloze task. Future work could explore whether coherence ratings for all clusters that were learned over the CLOC cloze dataset influence the model's accuracy in accounting for production times.

With these positive results, we now turn to the application of cluster-based measures that allow us to quantify the semantic predictability of words in context. Specifically, in Experiment 2 we test whether the probability of a word's semantic cluster has a positive effect on production latencies and relate the results to theories of competitive lexical selection.

Experiment 2

The cloze task requires engaging the production system to select an appropriate word or response given the context. The speed with which people are able to respond in this task depends in part on the predictability of the final word. In speeded cloze tasks, naming latencies to produce a response increase as cloze probabilities decrease (Staub et al., 2015; Bannon, Gollan, & Ferreira, 2024), which has supported the proposal that cloze probability can act as a proxy for the degree of activation that a word has during the production process. Under most models of language production, words are selected in order to align with the intended message via a translation between a semantic representation and an articulatory sequence (Dell, 1986; Levelt, 1989; Degen, 2023). According to some accounts, the selection of words is competitive, in which all of the words that could be selected for production compete for activation, with greater competition leading to longer reaction times (see Spalek, Damian, & Bölte, 2013 for a review). Non-competitive accounts, however, state that words must simply reach a threshold for production (Mahon et al., 2007; Oppenheim, 2023) and the word that does so first is ultimately articulated. Evidence has been presented in favor of both competitive and non-competitive accounts, with task design and structure strongly affecting patterns of behavioral results (Spalek et al., 2013).

Most pertinent to the present study, the results of analyses of reaction times in the speeded cloze task of Staub et al. (2015) supported a non-competitive lexical selection process in cloze production. Specifically, while speeded naming latencies were correlated with cloze probabilities, such that high-probability words were produced with shorter reaction times, the probability of the next-most probable word was not a major predictor of reaction times, suggesting that it did not negatively impact a word's production to have a high-probability competitor. Follow-up analyses assessed the impact on naming speed of semantic relatedness, which was quantified using the latent semantic analysis (LSA) approach (Landauer & Dumais, 1997). While Staub et al. found that highly constraining sentences elicited responses that were generally semantically similar, there was little to no effect on participants' naming latencies of the semantic similarity between a participant's response and the modal response. That is, greater semantic similarity between the top response and other responses did not consistently influence response times. The authors argued that this pattern suggests that semantic competition is not a major factor in lexical production. These results from Staub et al. (2015) contradicted competitive selection models by Levelt et al. (1999), which predicted that a high degree of activation of other responses should lead to slower reaction times. However, Staub et al.'s analyses only compared the (context-insensitive) word vectors between a response and the most frequent response for a given preamble; this analysis discards the similarity of a response to other continuations and lacks contextual information about a word's role in a sentence. Thus, it remains possible that some level of competition does contribute to selection during production, which was undetected in previous work.

Another concern about the Staub et al. (2015) task is that, similar to common practice in many laboratory tasks, it was designed to test predictability effects only for sentence-final productions (c.f., Federmeier et al., 2007; Peelle et al., 2020). Sentence-final words may not be processed or produced the same way as other words throughout sentences; for instance, processes may be engaged to "wrap up" the representation of the sentence and prepare for the next utterance (Just & Carpenter, 1980; Kuperberg et al., 2011). It is therefore unclear how much the results of Staub et al. (2015) are task- and stimulus-specific. Additionally, very little is known about production in the serial cloze task, despite the task's prevalent use in comprehension-oriented psycholinguistics. Many datasets (e.g., Luke & Christianson, 2016; Lowder et al., 2017) were gathered through untimed web surveys, which do not include latency information. Thus, to understand the serial cloze task and assess how much semantic competition affects language production speed, Experiment 2 analyzes the production latencies in the serial cloze dataset of Jacobs et al. (2022), which was built on the CLOC stimuli (Federmeier et al., 2007) for all words at all positions throughout a sentence.

Experiment 2 aims to leverage cluster information over cloze responses to produce a measure of semantic similarity that sheds light on the debate between competitive and non-competitive models of lexical selection. It expands on the LSA analyses of Staub et al. (2015)

and uses the BGMM clustering technique to probe the question of the impact of semantic competitors on production speed. In contrast to the cosine similarity analyses of Staub et al., we approximated whether a word has semantic competitors for a given cloze preamble by identifying whether a response is a singleton within a semantic cluster, or whether it is included with other responses in a larger semantic cluster. Additionally, our approach allowed us to model competition as a continuous variable, by including the cloze probability of a word at a particular position, as well as the cluster probability of the word's associated cluster. As stated previously, competitive theories of lexical selection (Levelt et al., 1999) propose that a sufficiently high degree of activation of a response's semantic neighbor should slow the production of that word. On the other hand, non-competitive accounts of lexical selection (e.g., Oppenheim, 2023) state that each response is activated independently of the other, and thus the presence of other words in a cluster would not impact production response times.

If semantic structure that is relevant to production in the cloze task can be captured by the BGMM, then competitive accounts predict that production latencies will be slower in non-singleton cases relative to words that stand alone in a semantic category for a given preamble, since the presence of other words in the cluster will lead to competition. Competitive accounts also predict a significant interaction between cloze probability and semantic competitor presence, such that high-probability responses with semantic competitors should be more difficult to produce than high-probability responses without them. On the other hand, non-competitive theories make the prediction that the presence of semantic neighbors within a cluster should not negatively impact production speeds. It is even possible that semantic neighbors could facilitate production, as activation speed could increase with greater semantic neighbors present; that is, if spreading activation of semantic features leads to facilitated production, than a greater number of semantic neighbors could produce higher semantic activation of the target word more rapidly. However, such a result would be unexpected: Prior work in language comprehension has shown that more distant semantic neighbors facilitate processing speed, while near neighbors can slow it down (Mirman & Magnuson, 2008).

Method

The serial cloze task data was obtained from Jacobs et al. (2022), which details the recruitment procedure and task structure. 158 participants whose first language was English were recruited through Prolific. The dataset consists of 109,225 completions of sentence fragments taken from the CLOC stimuli of Federmeier et al. (2007). In the experiment, participants viewed a sentence fragment word-by-word (e.g., "Most __", "Most of ___", ..., "Most of the trees died after last year's ___"), and they were told to guess the next word to fill in the blank for each word in the entire sentence. Participants never guessed the first word of the sentence. Due to the large number of responses required, participants provided annotations for an average of 703 unique preambles across 70 sentences. The task took approximately 1 hour. As the task was implemented for the web in jsPsych (De Leeuw, 2015), each participant's response in this serial cloze task is associated with a latency to complete the entire word, rather than the time to start typing the word; we refer to this measure as production time. We standardized participants' responses to normalize case and spelling and computed the cloze probability of each response (the proportion of participants who named that word for that preamble), as well as a cluster probability (the proportion of cloze responses for the cluster to which a given response belongs).

We define cloze probability and cluster probability as follows:

1) $$cloze\ probability = p(w \mid C) = \frac{count(response{=}w \cap context{=}C)}{count(context{=}C)}$$

2) $$cluster\ probability = p(cluster \mid C) = \frac{count(response \in cluster \cap context{=}C)}{count(context{=}C)}$$

Such a definition, computed over individual participant responses *i*, allows us to treat cloze responses as subsets of cluster responses, and thus produces a proper estimate of $p(w \mid C)$ as well. Table 6 provides a representative example of the cluster probability derivation for a single preamble.

Table 6. Cluster probabilities for a single preamble, "Most of the trees died after last year's _____"

Cluster ID	Cluster probability	Response	Cloze probability
21	0.697	fire	0.395
		drought	0.116
		storm	0.069
		hurricane	0.023
		harvest	0.023
		flood	0.023
		fires	0.023
		frost	0.023
32	0.116	winter	0.069
		summer	0.023
164	0.093	forest	0.093
11	0.046	cold	0.023
		heat	0.023
14	0.023	snow	0.023
126	0.023	ice	0.023

When we consider preambles that have at least two responses that belong to the same cluster, 14% of preamble-cluster pairs (3,486 of 25,073) meet this criterion. At the word level, 10,084 responses (of 31,671) belong to a cluster that has a competitor, which represents 32% of all response types. Cloze probabilities strongly correlate with word length (Pearson's ρ = -.21, $t(31669)$ =-37.40, $p < .001$), contextualized cluster probability (ρ = .74, $t(31669)$ = 195.84, $p < .001$), and log word frequency (ρ = .26, $t(31669)$ = 47.51, $p < .001$). Due to the way we compute cluster probabilities, which are derived from cloze probabilities, it is expected that there will be high correlations between the two. We address the issues with this strong collinearity by conducting forward stepwise model comparison analyses instead of relying solely on the significance of p values.

Results

We analyzed production times using a linear mixed effects model with random intercepts by participant, preamble, and the word that the participant produced. The production response time distribution is skewed somewhat, with a median production response time of 2.34 seconds and a mean response time of 3.61 seconds. We present the full distribution as Figure 9.

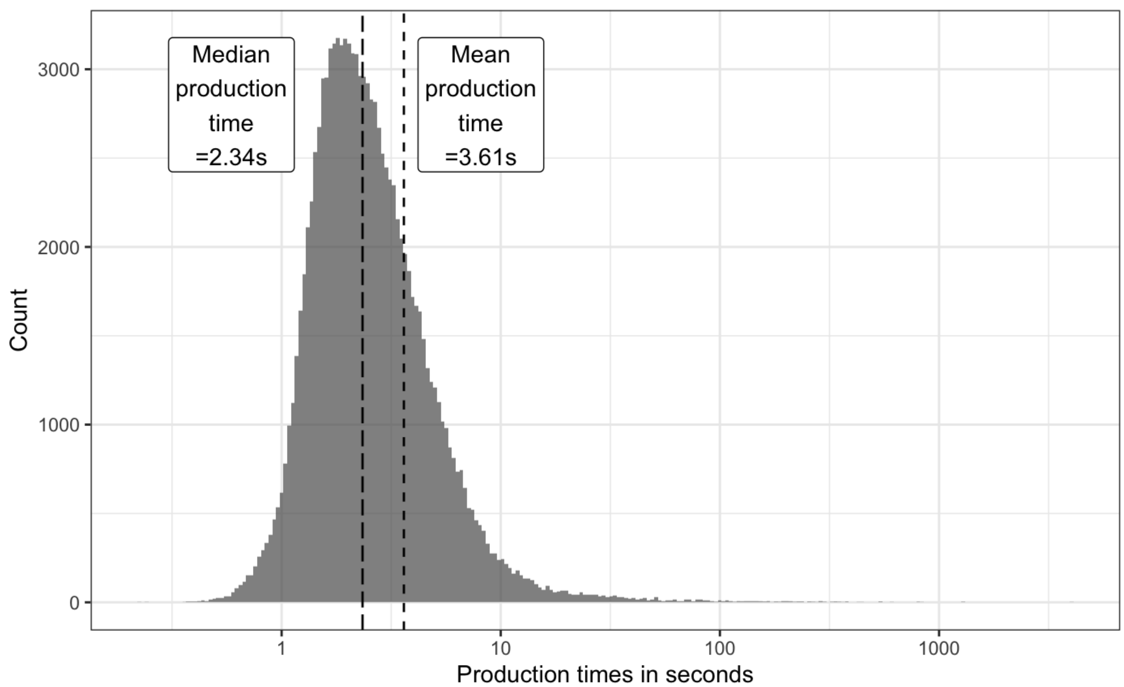

Figure 9. Production time distribution in seconds in the CLOC serial cloze data (Jacobs et al., 2022).

Our predictors of interest characterize the different production types. First, a participant's response is associated with a cloze probability of all participants producing that word as a response. Second, the word belongs to a cluster, which is associated with its own probability. Both probabilities were log transformed, centered and scaled with respect to the grand mean. Third, we created a binary predictor that encoded the presence or absence of other responses within the same semantic cluster, which we refer to here as the effect of *Competitors* on production times. *Competitors* captures the intuition that a competitor of any kind – rather than the specific value of the cloze or cluster probability – should lead to production costs. Similar analyses were conducted in Staub et al. (2015) by comparing singletons (cloze = 100%) to non-singletons in their analysis. We include word length in characters and smoothed log word frequency in the SUBTLEX-US corpus (Brysbaert & New, 2009) as control variables. Word frequency was smoothed by assuming a count of 1 (and thus log frequency of 0) for all responses not in SUBTLEX-US. We conducted model comparison analyses to determine whether including the probability-only (continuous) predictor was more appropriate to account for semantically-explained variance in production response times (in addition to the control predictors above), and whether additionally specifying a categorical predictor indicating the presence of a competitor would improve fit to the data beyond cluster probability.

We mean-centered and scaled the all predictors for analysis. Finally, we considered interactions between all probability variables and performed forward stepwise model comparison to evaluate the contribution of each predictor. Models were constructed using the R (version 4.3.1) package lmerTest (Kuznetsova et al., 2020; version 3.1-3). We used forward stepwise model comparison to determine the combination of predictors that was justified by the data and retained all interactions. We report the results of the final model from this analysis in Table 7 below. Further details of the lower-order models may be found at https://osf.io/n4avt/.

Table 7. Linear mixed effects model of log scaled production times.

Formula: RT (log transformed and scaled) ~ (1 | Preamble) + (1 | Response) + (1 | Participant) + Log Cloze Probability * Log Cluster Probability * Competitors + Word length + Word frequency

Fixed effect	Estimate	SE	*t*	*p*
Intercept	-0.003	0.04	-0.08	n.s.
Word length in characters	0.19	0.005	38.83	< .001
Log SUBTLEXUS frequency	-0.08	0.006	-13.64	< .001
Log cloze probability	-0.08	0.01	-7.25	< .001
Log cluster probability	-0.11	0.01	-10.79	< .001
Competitor present	0.02	0.004	4.93	< .001
Cloze probability * Cluster probability	-0.03	0.003	-10.47	< .001
Cloze probability * Competitor	-0.02	0.004	-3.62	< .001
Cloze * Cluster * Competitor	-0.01	0.002	-4.46	< .001

With respect to main effects in the model, the results of the best model replicated the previous finding in Staub et al. (2015) that higher-probability responses in the cloze task are produced significantly more quickly. Novel to the BGMM approach, we also find that words from high-probability clusters are produced more quickly, as supported by model comparison. At the same time, there was a significant main effect of Competitors, such that we observed slower production speeds when a participant's response did (versus did not) have a semantic competitor according to the BGMM. We plot the effect of competitors on production times in Figure 10.

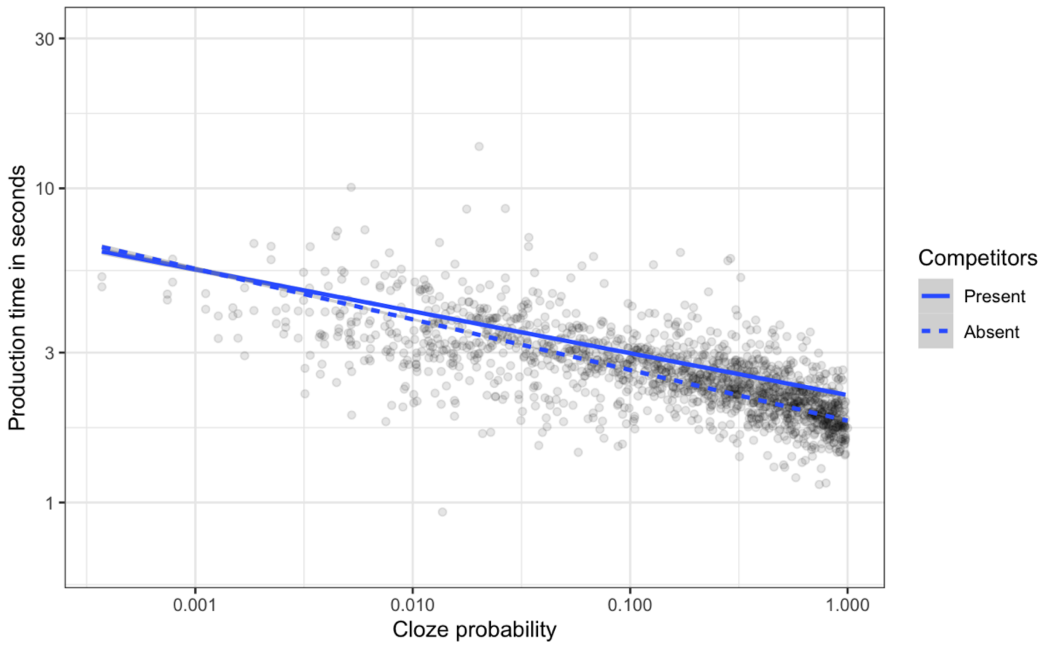

Figure 10. Relationship between cloze probability and production time by competitor presence. Competitors incur a cost to production times.

The final model also contained several interactions. There was a significant interaction between cloze probability and cluster probability, such that the effect of cloze probability on production times leads to even faster production when a response is the predominant or only word in its semantic cluster. That is, higher-probability responses within high-probability clusters are produced even faster than higher-probability responses in lower-probability clusters. Thus, there is a critical interaction that captures the intuition that the effect of a word's lexical predictability is modulated by its semantic predictability. We visualize this relationship in Figure 11.

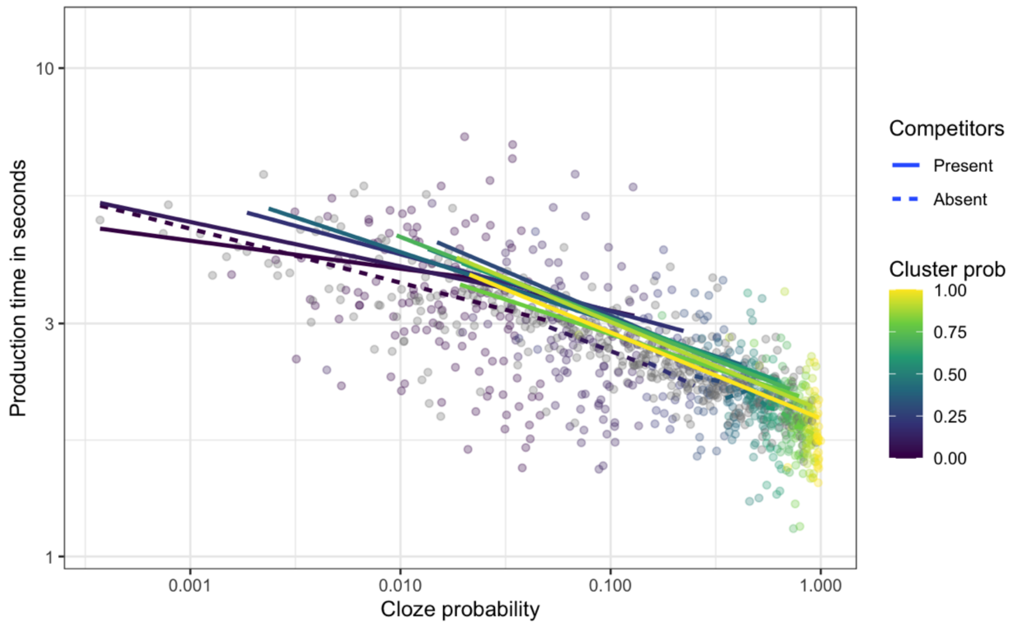

Figure 11. Relationship between cluster probability, cloze probability, and production times, with cluster probability binned for visualization. The slopes associated with high-probability clusters are steeper than slopes of low-probability clusters.

We found evidence for two additional interactions. Next, we observed an interaction between cloze probability and competitor presence. The effect of cloze probability on production times is slightly more negative (steeper) when competitors are not present compared to when they are absent (Figure 10). Finally, we observed a three-way interaction between cloze probability, cluster probability, and competitor presence, such that words with high cloze probability and high cluster probability were produced even faster in the absence of a competitor (e.g., when a word was the only one in its category). Specifically, when a word is the only response within a cluster, production response times are the fastest, replicating the singleton effect of Staub et al. (2015).

To better understand the interactions between cloze probability and cluster probability, we also visualized the relationship between the conditional probability of a response given a cluster (i.e., *p(w | C)* above), comparing across different bins of cluster probability (*C*). By comparing within a range of cluster probabilities and looking at the relative probability of a response within that cluster (thus, conditionalized on specific values), we can clearly see a cost to producing low-probability words even in very high-probability clusters, such that there are large costs to very low probability words and very low costs to extremely high-probability words within a cluster, in addition to main effects of cloze probability and cluster probability. That is, when a word is a singleton in its cluster, it is produced even more quickly than words with high, sub-1 cloze probability. Future work should explore the specific shape of this curve and its potential relationship to competitive accounts of selection. We present this visualization below in Figure 12.

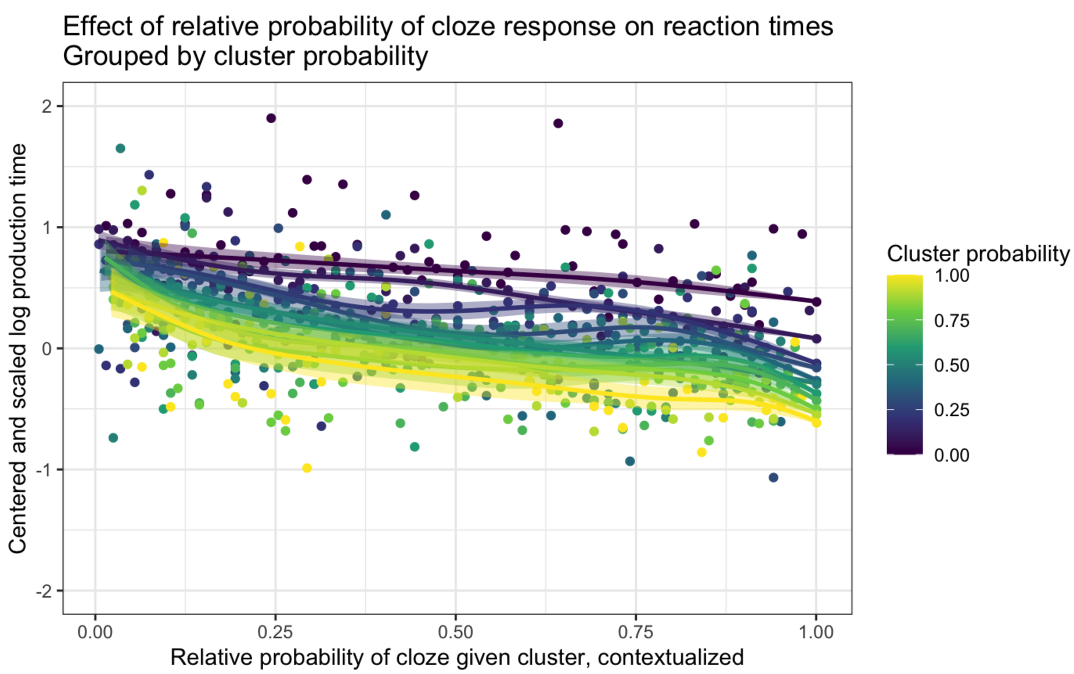

Figure 12. Relationship between cloze probability within a cluster and production response times for contextualized cluster probabilities.

Discussion

Taken together, the results of Experiment 2 provide some support for the presence of competition during lexical selection, at least for a serial cloze task with no time pressure. The significant main effect of the factor indicating the presence of semantic competitors as extracted from BGMM clusters shows that there is a cost to choosing between multiple alternatives. Words from high-probability clusters are nevertheless produced more quickly than those from low-probability clusters, suggesting that activation is passed first from semantics to words, and shared across a set of words that are likely to arise in a particular context. Most importantly for theories of semantic competition in language production, the results of the three-way interaction suggest that high-probability responses for a preamble that lacks any other responses in the same BGMM cluster are named much faster on average than high-probability responses with semantic competitors.

The discrepancy between the findings of Experiment 2 and the Staub et al. (2015) data may be due to task differences between speeded and non-speeded cloze tasks. However, while we cannot fully infer the response time distributions without their data, in their paper they produced figures with means in the range of 1400 milliseconds for lower-probability (slower) responses. To approximate this, we excluded reaction times above 2000 milliseconds from analysis and re-ran the previously described mixed effects model, retaining approximately 40% of the response times in our data. Including only these RTs did not qualitatively change our results – there were still major contributions of cluster probability to production speed, though the effect of Competitor was no longer significant or justified to include in the final model, suggesting that competition may arise later, potentially during deliberation and final response submission. We present these analyses in Appendix B.

Another potential difference between the Staub et al. findings and the present study could derive from the use of vectors from Latent Semantic Analysis for an analysis that compared the vector similarity between the top response and every other response. First, vectors derived from LSA, as used in Staub et al. (2015), rely on a count-based procedure, rather than a prediction-based procedure (Baroni et al., 2014). By contrast, the BGMM approach takes each cloze response from RoBERTa (Liu et al., 2019) – a "predict" model – and produces a context-sensitive representation that may be more sensitive to the micro-structure of semantics encoded at the message level (Levelt et al., 1999). Thus, a major difference could derive from a lack of contextual similarity and/or a difference in data sources. To test this idea, we conducted an additional analysis using uncontextualized vectors derived from the initial input embedding layer of RoBERTa. That is, different instances of the same word will always be assigned to the same cluster when using the first layer, because its representation does not incorporate context in any way. The results for analysis of the uncontextualized embeddings fully replicated the findings using the contextualized embeddings in Table 7. This suggests that the context-specific nature of the embeddings is not the primary contributor to discrepancies with the Staub et al. findings. We present these analyses in Appendix C.

Importantly for researchers attempting to study semantic processing in language production, the results of our BGMM-based cluster probability analyses show that the semantic structure of cloze responses can be quantified by clustering large language model-derived embedding representations, and that identifying semantic neighbors and/or competitors with this method can have predictive power for production times and naming latencies. We observe that, given a particular preamble, words that occur in clusters with other words are subject both to slowdown effects related to the presence of a competitor as well as facilitatory effects based on semantic (cluster-level) predictability. Being able to identify and characterize both of these effects is made possible by clustering large language model-derived representations of cloze responses, which may open new avenues for researchers hoping to understand semantic facilitation and competition in language production.

The results speak modestly to competitive versus non-competitive theories of lexical selection. We find evidence for competition, as words that were not sole members of a semantic cluster for a given preamble were slower to be named than words that were singletons. At the same time, there is clearly a positive effect of semantic neighbors as well. That is, words that belong to a high-probability cluster are also easier to name, and this is especially true for highly probable words within a cluster. The BGMM clustering method shows that semantic similarity and semantic competition in lexical selection arise via complex dynamics, with both competition and facilitation existing to different extents (Spalek et al., 2013). We concede that our analysis of production times does not directly speak to more standard computational accounts of competitive lexical selection that were primarily meant to explain short-term language production dynamics, such as those that might be used in the production of spontaneous utterances or selection of a name for a picture. For instance, speeded cloze tasks such as Staub et al. (2015) do not demonstrate clear effects of semantic competition on naming. This could be due to the procedure (speeded cloze), the type of stimuli (sentence-final responses only), the modality and measurement (speech onset latencies), or due to the use of similarity-based measures for quantifying the contribution of semantics to production, rather than cluster probabilities. In the unspeeded serial cloze data from Jacobs et al. (2022), production times may reflect many stages of deliberation prior to responding, and therefore may reflect revision processes as well as lexical access. Future studies could enforce a stricter timeline in the serial cloze task to make the task more comparable to Staub et al. (2015), or assess sentence-final cloze responses to the Federmeier et al. (2007) stimuli, and potentially yield different patterns of results from those that we report here.

General Discussion

A major challenge in working with human cloze responses and relating measures that are derived from them to human language processing dynamics is in characterizing the semantic structure of the responses that participants make. For instance, the probability of a set of upcoming words belonging to a particular semantic cluster may be high, but the probability of a specific word from that cluster may be low. Understanding how much comprehension and production depend on high-level properties such as the degree of a word's syntactic or semantic fit to the context has been challenging to quantify in an automated way. This work demonstrates that it is possible to quantify higher-order semantic structure using clustering models build on large language model representations and that the resulting clusters are useful for psycholinguistic applications, such as modeling the contribution of semantics to language production. In this work, we leveraged a large language model (RoBERTa; Liu et al., 2019) to obtain approximated semantic representations of words that were provided as responses in several cloze datasets (Luke & Christianson, 2016, 2018; Lowder et al., 2017; Jacobs et al., 2022).

By training Bayesian Gaussian Mixture models (BGMMs), we were able to characterize the semantic space of participants' responses. In Experiment 1, we showed that the clusters the models produce are both qualitatively (Experiment 1A) and quantitatively (Experiment 1B) coherent. Clusters that have high degrees of semantic coherence also tend to be taxonomically coherent. The clusters themselves also provide predictive power for examining how the probability of a semantically-defined cluster of possible lexical items, as well as the presence of a competing lexical item within that cluster, affect production latencies in a serial cloze task (Jacobs et al., 2022). In Experiment 2, therefore, we show that the BGMMs trained on large language model representations capture semantic structure in a manner that matters for human processing and that can provide novel insights into language processing.

The approach of the present study presents several advances beyond previous approaches. Firstly, many previous studies have demonstrated a tight relationship between cloze probabilities and processing speed, but studies of the semantic structure of cloze responses have been relatively limited, generally reflecting word-to-word similarity or word-to-context similarity scores (Luke & Christianson, 2016; Staub et al., 2015; Smith & Levy, 2011) rather than clusters whose contents can be easily inspected. The BGMM approach produces inspectable semantic clusters that, despite the data-driven nature of the approach, we found to be taxonomically coherent based on human-provided ratings of coherence. Second, our approach operationalizes semantic competition not by the average distance between a word and (some set of) other responses, but as the presence of words in the same semantic cluster for a given preamble. That is, the possibility of competition – and its consequent costs to production – arises only when other words in a similar category are also valid responses to the semantic space delineated by the context in the view of a particular individual. This approach not only potentially more closely aligns with how competition naturally arises, but also can be expanded to provide more fine-grained detail – for instance, distances between words in a cluster, or between clusters, can be calculated and added as factors into models to tease apart different aspects of competition during language production.

The current study largely focused on language *production*; however, we envision this method being broadly applicable across psycholinguistics, including language comprehension. More specifically, in the current study, our novel clustering method was incorporated into statistical analyses predicting production times to provide unique insight into competitive activation during language processing. In future work, the method can also be applied to explain variability and elucidate the mechanisms underlying neural signatures of language comprehension. For instance, the N400 component of the event-related potential (ERP) is elicited following meaningful stimuli, and the amplitude of this response is reduced following semantically congruous (Lau et al., 2013) or predictable words (Federmeier et al., 2007) compared to incongruous or unpredictable words. Interestingly, recent work has demonstrated that words that are less predictable than highly predictable sentence endings still elicit facilitated N400 responses compared to unpredictable words (Frade et al., 2022; Szewczyk & Federmeier, 2022), and elicit N400s similar in magnitude to highly predictable endings when the words are semantically related to the highly predictable endings (Brothers et al., 2023). This work supports the possibility that predictive processing during language comprehension may not be competitive or selective for specific lexical items, but rather may depend on broader semantic pre-activation (see review in Federmeier, 2022). However, the aforementioned studies only used specific words that were either highly predictable or less predictable, and more standard measures of semantic similarity, as is common in language processing studies. It is possible that the semantic pre-activation occurring during prediction is focused into specific semantic clusters, such that words that are within the same cluster may elicit similar N400 responses, whereas words from different clusters may differ in their N400 response, which we can identify and directly test with our method. Additionally, words that are unpredictable given a particular context may also elicit post-N400 frontal positivity or posterior positivity responses, depending on the plausibility of the word given the context (DeLong et al., 2014). Although these responses may reflect some sort of revision processes (Kuperberg et al., 2020), the exact mechanism of these responses remains largely unknown. One possibility is that activation of a plausible word outside of the cluster containing the most predictable word requires recruitment of additional resources, leading to elicitation of post-N400 responses. Thus, in future work we can potentially provide novel insight into the neural mechanisms of prediction and revision using our method.

The approach we outline here also provides a means to understand the internal structure of large language models as well as to cluster language model guesses about next words to better understand the output. To our knowledge, no large language model to date assigns probabilities to next words that perfectly align with human cloze probabilities. However, cloze data for some tasks may be costly to collect and may be incomplete relative to a very large vocabulary, making language models potentially useful for estimating the probabilities of responses that people would never provide in a cloze task (Szewczyk & Federmeier, 2022). Using our approach, researchers can take the top k guesses produced by a large language model, or generate a large number of continuations for a given preamble, and cluster the embeddings associated with these outputs to get a sense of which clusters the model guesses and how probable these different semantic clusters are. We anticipate that our method will open up avenues that can help researchers design techniques to reshape language models to produce more human-like continuations in cloze tasks, which will improve their psychometric fit (Eisape et al., 2020).

The Bayesian Gaussian mixture modeling approach here also extends our understanding of semantic processing to a formal, computational-level account level. By using predictive inference to learn semantic clusters from data, we are able to arrive at a sense of semantic similarity for words within the same cluster, versus different clusters. Historically, measures such as cosine similarity have been used to compare linguistic material in tasks such as semantic priming in lexical decision tasks (Landauer, Foltz, & Laham, 1998) or the similarity of a cloze response to its context (Luke & Christianson, 2016) or to other responses (Staub et al., 2015). Gaussian mixtures are able to provide the relevant comparison set on the basis of probabilistic comparison, assigning cloze responses to clusters proportionate to their probabilistic distinctiveness from other data points. Researchers may apply this approach to help them specify the comparison set for their analyses in a data-driven way.

A number of empirical questions remain about the BGMM approach. First, it is currently unclear exactly how many clusters to learn for a given cloze dataset, and how to empirically determine the optimal number of clusters. While the clusters used in our study were coherent and well-formed within the parameter space that we explored, it is possible that using a slightly different number of clusters or changing the clustering parameters to some degree would have produced clusters with even higher coherence or explanatory power. We determined the weight concentration prior by visual inspection and others may find a more quantitative approach better suited to their questions. We wish to note that while one can assess GMM models of different complexity by comparing model likelihoods, building models with more clusters will typically lead to a statistically "better" model of cloze productions without any obvious gains in coherence. Thus, in these cases, we recommend conducting rating studies of cluster coherence and assessing the linguistic structure of the clusters (e.g., syntactic uniformity) in lieu of model-based statistical evaluation. Second, some types of model representations from large language models may be better-suited to modeling cloze data; although we focus on RoBERTa (Liu et al., 2019), other large language models, such as GPT-2 and Pythia may show a tighter correspondence to cloze data but may vary in what kinds of semantic information they encode. For instance, multimodal representations may better reflect semantic knowledge than text-only representations (Merkx et al., 2023). Even within a single language model, some layers may be more useful for different tasks (Tenney et al., 2019), or all layers may be useful for modeling linguistic knowledge (Charpentier & Samuel, 2023). However, we believe the general parameter range that we have tested represents a good starting point for researchers who wish to cluster their own cloze or production data, especially if using similar LLM architectures.

Bayesian Gaussian mixture models are not the only clustering framework that might prove useful for identifying semantic neighbors or competitors. For example, algorithms such as DBSCAN (Ester et al., 1996) have shown some success in capturing the semantic structure of human psycholinguistic datasets (Wilson & Marantz, 2022). Other clustering models, such as k-means (MacQueen, 1967) may also permit repeated data points, as the definition of a cluster is defined over distances to centroids. Additionally, the BGMM implementation that we used here relies on k-means to determine the initial hypotheses about where cluster means are located; as such, optimizing *k* explicitly will be pursued in future work. While BGMMs have specific advantages, which we summarize in the Introduction, we encourage others apply different clustering algorithms to embeddings of cloze responses that produce similarly useful representations that will improve cognitive modeling of semantic processing.

In conclusion, we have developed a novel approach that leverages LLMs and BGMMs to group the cloze task responses produced by participants into distinct semantic clusters, which can be used to investigate the mechanisms of semantic predictability during language processing. We validated this approach with a coherence-rating task, and used the metrics provided by our method to predict cloze production times and test theories of competitive activation during language production in new ways. We have provided code that allow researchers to use this approach with their own data and hope the broader psycholinguistic community will find this method useful in their own studies of language processing.

References

Asr, F. T., Zinkov, R., & Jones, M. (2018, June). Querying word embeddings for similarity and relatedness. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)* (pp. 675-684).

Bahdanau, D., Cho, K. H., & Bengio, Y. (2015, January). Neural machine translation by jointly learning to align and translate. In *3rd International Conference on Learning Representations, ICLR 2015*

Bannon, J., Gollan, T. H., & Ferreira, V. S. (2024). Is predicting during language processing worth it? Effects of cloze probability and semantic similarity on failed predictions. *Journal of Experimental Psychology: Learning, Memory, and Cognition.* Advance online publication. https://doi.org/10.1037/xlm0001347

Baroni, M., Dinu, G., & Kruszewski, G. (2014, June). Don't count, predict! a systematic comparison of context-counting vs. context-predicting semantic vectors. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 238-247).

Beekhuizen, B., Armstrong, B. C., & Stevenson, S. (2021). Probing lexical ambiguity: Word vectors encode number and relatedness of senses. *Cognitive Science*, *45*, e12943.

Bishop, C.M. (2006). *Pattern Recognition and Machine Learning* (Information Science and Statistics).

Brothers, T., Morgan, E., Yacovone, A., & Kuperberg, G. (2023). Multiple predictions during language comprehension: Friends, foes, or indifferent companions? *Cognition*, 241, 105602.

Brunila, M., & LaViolette, J. (2022, July). What company do words keep? Revisiting the distributional semantics of JR Firth & Zellig Harris. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 4403-4417).

Brysbaert, M., New, B., & Keuleers, E. (2012). Adding part-of-speech information to the SUBTLEX-US word frequencies. *Behavior Research Methods*, *44*, 991-997.

Charpentier, L. G. G., & Samuel, D. (2023, December). Not all layers are equally as important: Every Layer Counts BERT. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning* (pp. 238-252).

Christensen, R. H. B. (2023). ordinal: Regression Models for Ordinal Data. doi: https://doi.org/10.32614/CRAN.package.ordinal

Chronis, G., & Erk, K. (2020, November). When is a bishop not like a rook? When it's like a rabbi! Multi-prototype BERT embeddings for estimating semantic relationships. In *Proceedings of the 24th Conference on Computational Natural Language Learning* (pp. 227-244).

De Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a Web browser. *Behavior research methods*, *47*, 1-12.

de Varda, A. G., Marelli, M., & Amenta, S. (2023). Cloze probability, predictability ratings, and computational estimates for 205 English sentences, aligned with existing EEG and reading time data. *Behavior Research Methods*, 1-24.

Degen, J. (2023). The rational speech act framework. *Annual Review of Linguistics*, *9*, 519-540.

Dell, G. S. (1986). A spreading-activation theory of retrieval in sentence production. *Psychological review*, *93*, 283.

Dell, G. S., & Chang, F. (2014). The P-chain: Relating sentence production and its disorders to comprehension and acquisition. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *369*(1634), 20120394.

DeLong, K. A., Quante, L., & Kutas, M. (2014). Predictability, plausibility, and two late ERP positivities during written sentence comprehension. *Neuropsychologia*, *61*, 150-162.

Devlin, J., Chang, M-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1 (Long and Short Papers), pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Eisape, T., Zaslavsky, N., & Levy, R. (2020). Cloze distillation: Improving neural language models with human next-word prediction. Association for Computational Linguistics (ACL).

Ester, M, Kriegel, H-P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the 2nd ACM International Conference on Knowledge Discovery and Data Mining* (KDD). 226–231.

Ethayarajh, K. (2019, November). How Contextual are Contextualized Word Representations? Comparing the Geometry of BERT, ELMo, and GPT-2 Embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 55-65).

Federmeier, K. D. (2022). Connecting and considering: Electrophysiology provides insights into comprehension. *Psychophysiology, 59,* e13940. https://doi.org/10.1111/psyp.13940

Federmeier, K. D., Wlotko, E. W., De Ochoa-Dewald, E., & Kutas, M. (2007). Multiple effects of sentential constraint on word processing. *Brain Research*, *1146*, 75-84.

Frade, S., Pinheiro, A. P., Santi, A., & Raposo, A. (2022). Is second best good enough? An EEG study on the effects of word expectancy in sentence comprehension. *Language, Cognition and Neuroscience*, 37, 209-223.

Fraley, C., Raftery, A. E., Scrucca, L., Murphy, T. B., Fop, M. (2024). mclust: Gaussian Mixture Modelling for Model-Based Clustering, Classification, and Density Estimation. doi: 10.32614/CRAN.package.mclust

Grand, G., Blank, I. A., Pereira, F., & Fedorenko, E. (2022). Semantic projection recovers rich human knowledge of multiple object features from word embeddings. *Nature Human Behaviour*, *6*, 975-987.

Griffiths, T. L., Steyvers, M., & Tenenbaum, J. B. (2007). Topics in semantic representation. *Psychological Review, 114*, 211–244.

Jacobs, C. L., Hubbard, R. J., & Federmeier, K. D. (2022, February). Masked language models directly encode linguistic uncertainty. In *Proceedings of the Society for Computation in Linguistics 2022* (pp. 225-228).

Jacobs, C. L., & MacDonald, M. C. (2023). A chimpanzee by any other name: The contributions of utterance context and information density on word choice. *Cognition*, *230*, 105265.

Jacobs, C. L., & MacDonald, M. C. (2024). Constraint satisfaction in large language models. *Language, Cognition and Neuroscience*, 1231-1248.

Just, M. A., & Carpenter, P. A. (1980). A theory of reading: from eye fixations to comprehension. *Psychological review*, *87*, 329.

Kumar, A. A. (2021). *Modeling semantic structure and spreading activation in retrieval tasks* (Doctoral dissertation, Washington University in St. Louis).

Kuperberg, G. R., Brothers, T., & Wlotko, E. W. (2020). A tale of two positivities and the N400: Distinct neural signatures are evoked by confirmed and violated predictions at different levels of representation. *Journal of Cognitive Neuroscience*, 32, 12-35.

Kuperberg, G. R., Paczynski, M., & Ditman, T. (2011). Establishing causal coherence across sentences: An ERP study. *Journal of Cognitive Neuroscience*, *23*, 1230-1246.

Kutas, M., & Hillyard, S. A. (1980). Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science*, *207*, 203-205.

Kutas, M., & Hillyard, S. A. (1984). Brain potentials during reading reflect word expectancy and semantic association. *Nature*, *307*, 161-163.

Kuznetsova, A., Brockhoff, P. B., Christensen, R. H. B, & Jensen, S. P. (2020). lmerTest: Tests in linear mixed effects models. doi: 10.32614/CRAN.package.lmerTest

Landauer, T. K., & Dumais, S. T. (1997). A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction, and representation of knowledge. *Psychological review*, *104*, 211.

Landauer, T. K., Foltz, P. W., & Laham, D. (1998). An introduction to latent semantic analysis. *Discourse processes*, *25*, 259-284.

Lau, E. F., Holcomb, P. J., & Kuperberg, G. R. (2013). Dissociating N400 effects of prediction from association in single-word contexts. *Journal of Cognitive Neuroscience*, 25, 484-502.

Lau, J. H., Armendariz, C., Lappin, S., Purver, M., & Shu, C. (2020). How furiously can colorless green ideas sleep? sentence acceptability in context. *Transactions of the Association for Computational Linguistics*, *8*, 296-310.

Levelt, W. J. M. (1989). *Speaking : from intention to articulation*. MIT Press.

Levelt, W. J., Roelofs, A., & Meyer, A. S. (1999). A theory of lexical access in speech production. *Behavioral and Brain Sciences*, *22*, 1-38

Li, J., & Joanisse, M. F. (2021). Word senses as clusters of meaning modulations: A computational model of polysemy. *Cognitive Science*, *45*, e12955.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., ... & Stoyanov, V. (2019). RoBERTa: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.

Lowder, M. W., Choi, W., Ferreira, F., & Henderson, J. M. (2018). Lexical predictability during natural reading: Effects of surprisal and entropy reduction. *Cognitive Science*, *42*, 1166-1183.

Luke, S. G., & Christianson, K. (2016). Limits on lexical prediction during reading. *Cognitive Psychology*, *88*, 22-60.

Luke, S. G., & Christianson, K. (2018). The Provo Corpus: A large eye-tracking corpus with predictability norms. *Behavior Research Methods*, *50*, 826-833.

MacQueen, J. (1967, June). Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, No. 14, pp. 281-297).

Mahon, B. Z., Costa, A., Peterson, R., Vargas, K. A., & Caramazza, A. (2007). Lexical selection is not by competition: a reinterpretation of semantic interference and facilitation effects in the picture-word interference paradigm. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *33*(3), 503.

McLachlan, G. J., & Basford, K. E. (1988). *Mixture models: Inference and applications to clustering* (Vol. 38). New York: M. Dekker.

Merkx, D., Scholten, S., Frank, S. L., Ernestus, M., & Scharenborg, O. (2023). Modelling human word learning and recognition using visually grounded speech. *Cognitive Computation*, *15*, 272-288.

Merrill, W., Weiss, G., Goldberg, Y., Schwartz, R., Smith, N. A., & Yahav, E. (2020, July). A Formal Hierarchy of RNN Architectures. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 443-459).

Miaschi, A., Brunato, D., Dell'Orletta, F., & Venturi, G. (2020, December). Linguistic Profiling of a Neural Language Model. In *Proceedings of the 28th International Conference on Computational Linguistics* (pp. 745-756).

Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, *26*.

Miller, G. A. (1995). WordNet: a lexical database for English. *Communications of the ACM*, *38*, 39-41.

Mirman, D., Landrigan, J. F., & Britt, A. E. (2017). Taxonomic and thematic semantic systems. *Psychological Bulletin*, *143*, 499.

Mirman, D., & Magnuson, J. S. (2008). Attractor dynamics and semantic neighborhood density: processing is slowed by near neighbors and speeded by distant neighbors. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *34*, 65.

Mosbach, M., Degaetano-Ortlieb, S., Krielke, M. P., Abdullah, B. M., & Klakow, D. (2020, December). A Closer Look at Linguistic Knowledge in Masked Language Models: The Case of Relative Clauses in American English. In *Proceedings of the 28th International Conference on Computational Linguistics* (pp. 771-787).

Oh, B. D. (2021, June). Team Ohio State at CMCL 2021 shared task: Fine-tuned RoBERTa for eye-tracking data prediction. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics* (pp. 97-101).

Oppenheim, G. M. (2024, January 16). Lexical selection in language production. https://doi.org/10.31234/osf.io/k2s3p

Payne, B. R., & Federmeier, K. D. (2017). Pace yourself: Intraindividual variability in context use revealed by self-paced event-related brain potentials. *Journal of Cognitive Neuroscience*, *29*, 837-854.

Peelle, J. E., Miller, R. L., Rogers, C. S., Spehar, B., Sommers, M. S., & Van Engen, K. J. (2020). Completion norms for 3085 English sentence contexts. *Behavior Research Methods*, *52*, 1795-1799.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI blog*, *1*(8), 9.

Rasmussen, C. (1999). The infinite Gaussian mixture model. *Advances in Neural Information Processing Systems*, *12*.

Reynolds, D. A. (2009). Gaussian mixture models. Encyclopedia of biometrics, *741*, 659-663.

Roberts, S. J., Husmeier, D., Rezek, I., & Penny, W. (1998). Bayesian approaches to Gaussian mixture modeling. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *20*, 1133-1142.

Rogers, A., Kovaleva, O., & Rumshisky, A. (2021). A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics*, *8*, 842-866.

Rogers, T. T., & McClelland, J. L. (2004). *Semantic cognition: A parallel distributed processing approach*. MIT press.

Roland, D., Yun, H., Koenig, J. P., & Mauner, G. (2012). Semantic similarity, predictability, and models of sentence processing. *Cognition*, *122*, 267-279

Rothe, S., & Schütze, H. (2016, August). Word embedding calculus in meaningful ultradense subspaces. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (pp. 512-517).

Shain, C., Meister, C., Pimentel, T., Cotterell, R., & Levy, R. (2024). Large-scale evidence for logarithmic effects of word predictability on reading time. *Proceedings of the National Academy of Sciences*, *121*, e2307876121.

Smith, N., & Levy, R. (2011). Cloze but no cigar: The complex relationship between cloze, corpus, and subjective probabilities in language processing. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 33, No. 33).

Spalek, K., Damian, M. F., & Bölte, J. (2013). Is lexical selection in spoken word production competitive? Introduction to the special issue on lexical competition in language production. *Language and Cognitive Processes*, *28*, 597-614.

Sridhar, V. K. R. (2015, June). Unsupervised topic modeling for short texts using distributed representations of words. In *Proceedings of the 1st Workshop on Vector Space Modeling for Natural Language Processing* (pp. 192-200).

Staub, A., Grant, M., Astheimer, L., & Cohen, A. (2015). The influence of cloze probability and item constraint on cloze task response time. *Journal of Memory and Language*, *82*, 1-17.

Steyvers, M., Griffiths, T. L., & Dennis, S. (2006). Probabilistic inference in human semantic memory. *Trends in Cognitive Sciences*, *10*, 327-334.

Szewczyk, J. M., & Federmeier, K. D. (2022). Context-based facilitation of semantic access follows both logarithmic and linear functions of stimulus probability. *Journal of Memory and Language*, *123*, 104311.

Taylor, W. L. (1953). “Cloze procedure”: A new tool for measuring readability. *Journalism Quarterly*, *30*, 415-433.

Tenney, I., Das, D., & Pavlick, E. (2019, July). BERT Rediscovers the Classical NLP Pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 4593-4601).

Toscano, J. C., & McMurray, B. (2010). Cue integration with categories: Weighting acoustic cues in speech using unsupervised learning and distributional statistics. *Cognitive science*, *34*, 434-464.

Tuinman, J. J., Blanton, W. E., & Gray, G. (1975). The cloze procedure: An analysis of response distributions. *The Journal of General Psychology*, *92*, 177-185.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

Weiss, G., Goldberg, Y., & Yahav, E. (2021). Thinking Like Transformers. *Proceedings of the 38th International Conference on Machine Learning*, in *Proceedings of Machine Learning Research, 139*, 11080-11090.

Wilson, K., & Marantz, A. (2022, December). Contextual embeddings can distinguish homonymy from polysemy in a human-like way. In *Proceedings of the 5th International Conference on Natural Language and Speech Processing (ICNLSP 2022)* (pp. 144-155).

Wlotko, E. W., & Federmeier, K. D. (2012). Age-related changes in the impact of contextual strength on multiple aspects of sentence comprehension. *Psychophysiology*, *49*, 770-785.

Appendix A

Experiment 1 Cluster Coherence Rating Instructions

In this experiment, we will ask you to rate how well a set of words "goes together" (is similar, coheres). Some of these word groupings will feel more coherent than others, and they may also contain words that are similar to each other in different ways.

We will first ask you to give your overall impression of how well the words fit together. We will then ask you to assess two different "types" of similarity.

The first type of similarity is what we are here calling "categorical" similarity (it can also be called "taxonomic" similarity). This is describing the kind of similarity that is shared by members of a given category, like nouns naming mammals or kitchen utensils or verbs that refer to ways of controlling some kind of vehicle. When a set of words has high categorical similarity, a comparison of each of the pairs of words in a set will show that most pairs can be thought of as belonging to a shared category. Here is an example of a group of words with high taxonomic similarity because they all belong to the category of "fruit":

<u>Example 1</u>

- Apple
- Orange
- Banana
- Pomegranate
- Fig
- Cherry

Some categories are, of course, more general than others. For example, a group of words that all described edible items also has some level of categorical similarity. And even a set of things that all belong to the same part of speech, such as nouns, verbs, or adjectives, has a bit of categorical similarity, even if they do not mean similar things because, for example, they all are words used to describe something.

Thus, when assessing categorical similarity, you may want to first consider how many of the words in the set are the same part of speech (e.g., noun, verb, etc.). If many are from the same part of speech, next consider: (1) Are the words in the group all the same "kind" of thing; could they substitute for one another in some context? (2) How easy is it to find a category label to describe the words that are in this set?

If you feel that the statement (1) above is very true of a group of words, then you should give that group a high rating for categorical similarity.

On the other hand, if in considering (2) you notice that there isn't a good, existing category label that can apply to many of the words in the group, then the group might not show much categorical similarity. Instead, it is possible that that group may have thematic similarity, as described next, or no similarity at all.

Thematic similarity is another way that groups of words can be coherent. However, this type of similarity does not have specific categories that apply. Instead, this kind of similarity is characterized by items and actions that are likely to occur together in the description of an event or situation (for example, a painter buying art supplies from a craft store). The word "painter" refers to a human being and "art supplies" refers to tools used for making art. These do not belong to the same category of objects, so they don’t have categorical similarity. However, when they reference things that occur together in real-world scenarios and in sentences describing those scenarios, they show thematic similarity instead of categorical similarity.

When the words in a group occur together in real-world scenarios or when they can co-occur in a sentence, they show thematic similarity instead of taxonomic similarity.

Example 2

Here is a list that shows thematic similarity:

- artist
- purchase
- expensive
- pigments
- brush

To evaluate thematic similarity, try to visualize whether an event could be easily described using the group of the words that you see. If they can easily be turned into a commonplace event, then they probably demonstrate thematic similarity.

Experiment 1 Cluster Coherence Rating Debriefing

The study you participated in today asked you to evaluate the cohesiveness of clusters of words. We created several clusters using natural language processing systems as well as several "scrambled" clusters. Approximately half of the clusters you rated should have been pretty incoherent, while the other half were probably pretty reasonable, though not perfect.

We are using human judgments as part of a bigger project to understand what people predict during they read. We are attempting to quantify whether people guess words that are usually from one category (e.g., animals), situation (e.g., an artist visits an art store) or more unrelated. We think that you will mostly have rated the scrambled clusters lower in coherence than the algorithmically derived clusters.

We also asked you to rate whether clusters were more category-based than situation-based. We think that the NLP algorithms will generate more similarity for both category-based and situation-based measures. But, we do not know whether category-based or situation-based measures are more likely to show up in our models. So, wanted to see what you thought.

Thank you for participating! We appreciate the effort you went through. Please continue to be redirected to Prolific.

Appendix B

Analyses of sub-2000 ms production times to compare to Staub et al. (2015)

One potential discrepancy between the results of Experiment 2 and those of Staub et al. (2015) lies in the differences between the two production tasks and the related measures. Staub et al. measured speech onset latencies to name a cloze response, while the Jacobs et al. (2022) data contain whole-word typed production times. One way to attempt to equate the two experiments is to restrict the production times of the Experiment 2 data. Here, we report the results of such an analysis after restricting production times to be under 2000 milliseconds. The results are qualitatively similar to the full dataset, though some effects are attenuated in their effect size (cloze probability, word length, and word frequency). Other correlations to production times disappear in the restricted dataset and were not supported by model comparison: the effect of competitor presence; the interaction between cloze and competitor presence; and the interaction between cloze, cluster probability, and competitor presence. The results nevertheless generally support an interpretation that probable words and probable categories of words independently contribute to production times. Specifically, when high-probability words belong to high-probability categories, they appear to be produced even more quickly than when a high-probability word belongs to a lower-probability category, as seen in Experiment 2.

One potential interpretation of the lack of Competitor in the final model is that competition is only evident during slower, more deliberative production tasks. Future work should replicate the Jacobs et al. (2022) task with time pressure and collect first keystroke data to identify other sources of discrepancies.

Table B.1 Linear mixed effects model of log scaled production times.

Formula: RT (log transformed) ~ (1 | Preamble) + (1 | Response) + (1 | Participant) + Log Cloze Probability * Log Cluster Probability + Word length + Word frequency

Fixed effect	Estimate	SE	*t*	*p*
Intercept	-.08	0.08	-1.01	n.s.
Word length in characters	0.18	0.007	24.75	< .001
Log SUBTLEXUS frequency	-0.02	0.01	-1.95	= .05
Log cloze probability	-0.08	0.02	-5.00	< .001
Log cluster probability	-0.06	0.02	-3.89	< .001
Cloze probability * Cluster probability	-0.04	0.004	-8.15	< .001

Appendix C

Analyses of uncontextualized vs. contextualized cluster probabilities

Uncontextualized cluster probabilities (2) are derived in the same way way as contextualized cluster probabilities. However, a word's vector representation will always occur in the same cluster, no matter what linguistic preamble occurs beforehand, because the embeddings for that word do not incorporate knowledge of any of the other words in the sentence.

We define cloze probability and uncontextualized cluster probability as follows:

1) $$cloze\ probability = p(w \mid C) = \frac{count(response{=}w \cap context{=}C)}{count(context{=}C)}$$

2) $$cluster\ probability = p(cluster \mid C) = \frac{count(response \in cluster \cap context{=}C)}{count(context{=}C)}$$

The current analysis tests whether context sensitivity is necessary for competition to arise. As with contextualized cluster probabilities, cloze probabilities strongly uncontextualized cluster probability ($\rho = .68$, $t(31669) = 164.72$, $p < .001$), though less strongly so. A potential explanation for the stronger correlation between contextualized cluster probabilities relative to uncontextualized clusters is likely due to the tendency for the clustering model to learn clusters containing one or very few words, which appears to create, for any one preamble, more clusters that contain only a single cloze response.

Preambles do moderately influence the number of clusters. Almost definitionally, the number of clusters to which a string is assigned is expected to increase for a given word type because of the contextual representations. It is unclear, at first glance, whether the converse - the number of clusters associated with a preamble - should also increase when embeddings are contextualized. We compared the number of clusters for each preamble before and after clustering and then conducted a paired samples t-test showing that the number of clusters associated with a preamble was higher for contextual (mean 9.91 clusters) than uncontextual representations (mean 8.80); $t(2567) = -23.04$, $p < .001$.

Thus, the potential for competition is even more stark when considering uncontextualized embeddings; 47% (14,991 response types of 31,671) and 22% (5,534 of 25,073) of the preamble-cluster pairs reveal that highly similar words are produced. However, uncontextualized clusters that are inferred by the BGMM may not be able to capture competition, as in the LSA analyses presented in Staub et al. (2015). Thus, we conducted an analysis using the same formula as Table 7 in the main paper using uncontextualized cluster probabilities and present the results in Table C.1. We present a visualization of the three-way interaction in Figure C.1.

Table C.1 Linear mixed effects model of log scaled production times.

Formula: RT (log transformed and scaled) ~ (1 | Preamble) + (1 | Response) + (1 | Participant) + Log Cloze Probability * Log Cluster Probability * Competitors + Word length + Word frequency

Fixed effect	Estimate	SE	*t*	*p*
Intercept	0.0003	0.04	0.007	n.s.
Word length in characters	0.188	0.005	38.74	< .001
Log SUBTLEXUS frequency	-0.08	0.006	-13.19	< .001
Log cloze probability	-0.14	0.006	-23.10	< .001
Log cluster probability	-0.05	0.007	-8.38	< .001
Competitor present	0.003	0.004	0.73	n.s.
Cloze probability * Cluster probability	-0.03	0.003	-9.83	< .001
Cloze probability * Competitor	0.02	0.005	3.24	< .01
Cluster probability * Competitor	-0.02	0.005	-4.36	< .001
Cloze * Cluster * Competitor	-0.007	0.002	-3.22	< .01

Given this, it seems likely that applying context-specific probabilities, which increases the sparsity of the clusters, has the effect of affecting the relevance of competitors from different categories and overall reducing the presence of competitors within a specific preamble. Future work should determine the best factors to assign words to categories that are relevant for addressing questions of semantic neighborhood effects on production.

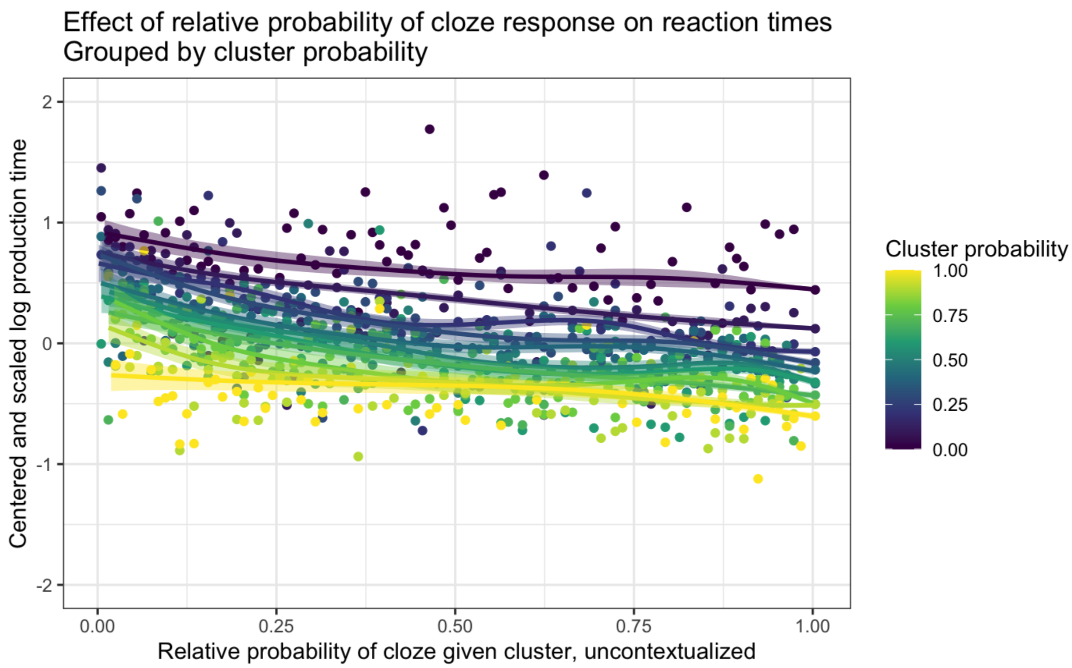

Figure C.1. Relationship between cloze probability within a cluster and production response times for uncontextualized cluster probabilities.